Increasing ratios in finite nonnegative linearization

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November 1, 2026

Abstract. For finite three-term recurrences normalized to have row sums one, the companion paper [5] proves nonnegative linearization (property (P)) when the ratio $\rho(i) = c_i/(b_i + c_i)$ of the downward coefficients to the profile $d_i = b_i + c_i$ is constant and the Newton coefficients of the profile are log-concave without internal zeros. We show that the ratio may increase with the index, provided the diagonal coefficients b_i are nondecreasing. The main result admits, on every such profile, every ratio whose Newton coefficients are log-concave without internal zeros, for instance exponential ratios. Separate theorems admit all convex quadratic ratios and, under the single inequality $r\sigma \le \frac{7}{4}t\tau$, the cubic ratios $r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$. The proofs resum the moves of the kernel recurrence of [5]. The key inequalities of the main result and of the cubic criterion are proved with exact computer certificates, and all results are formalized in Lean 4. Nondecreasing b alone, or nonnegative Newton coefficients of the ratio alone, do not suffice. We also give a self-contained proof, in the Newton basis, of Walkup's theorem that log-concavity of Newton coefficients is preserved under pointwise products.

Keywords. Finite orthogonal polynomials; nonnegative linearization; log-concave sequences; Newton series; computer-assisted proof.

MSC2020 Classification. 33C45; 42C05; 05A20.

1. Introduction

Szwarc [10, §5, Problem 2] asked for criteria, in the recurrence coefficients, for nonnegative linearization (property (P)) of finite systems of orthogonal polynomials. The companion paper [5] answers this for arrays whose diagonal and downward coefficients have a constant ratio: property (P) holds whenever the Newton coefficients of the profile are log-concave without internal zeros. This paper treats ratios $\rho(i) = c_i/(b_i + c_i)$ that increase with i.

Our main result admits every ratio whose Newton coefficients are log-concave, on every profile allowed in [5]. Here the Newton coefficients of a sequence (u_i) are $\Delta^k u(0)$, so that $u_n = \sum_k \binom{n}{k}\Delta^k u(0)$.

Theorem A *(Theorem 5.5).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros. Let $\rho(x) = \sum_k \rho_k\binom{x}{k}$, where $(\rho_k)_{k\ge 0}$ is nonneg-*

ative and log-concave without internal zeros and $\rho_0 > 0$. *Put* $a_i = 1 - d_i$, $b_i = (1 - \rho(i))d_i$ *and* $c_i = \rho(i)d_i$. *If* b *is nondecreasing, then the system has nonnegative linearization, including the products beyond the terminal degree.*

It admits, for example, the exponential ratios $\rho(x) = r(1+\varepsilon)^x$ on every such profile. For constant ρ it is the criterion of [5]. Its proof reduces a key inequality to symbolic coefficient certificates and finitely many exact checks, all verified in Lean.

Section 2 recalls the setting and the log-concavity inequality on which all proofs rest. Section 3 extends the kernel recurrence of [5] to arrays with independent diagonal and downward coefficients and proves property (P) when the Newton coefficients of c are log-concave without internal zeros and those of b are those of c times nonincreasing nonnegative weights (Theorem 3.1 and Corollary 3.2), and for arrays that are quadratic in the index (Proposition 3.3). In general some moves of the recurrence have negative weight; Section 4 handles them by a resummation, for ratios that increase linearly (Theorem 4.1) or as a convex quadratic (Theorem 4.2) on every log-concave Newton profile, under the condition that b be nondecreasing. The same monotonicity of the diagonal coefficients appears in the criteria of Askey and Szwarc for infinite systems [10, Theorems 3.1 and 3.2] and in the inequality criteria for finite systems in [6]. Section 5 treats ratios of higher degree: on profiles of Newton degree at most two (Theorem 5.1), and on every log-concave Newton profile when the Newton coefficients of the ratio are log-concave (Theorem 5.5) or, for cubic ratios, when $r\sigma \le \frac{7}{4}t\tau$ (Theorem 5.6 and Corollary 5.7). Proposition 5.3, due to Walkup [12], shows that log-concavity of Newton coefficients is preserved under pointwise products. Section 6 states the open problems; Appendix A records the exact counterexamples and exploratory tests.

The key inequalities of Theorems 5.5 and 5.6 and of Corollary 5.7, which gives the stronger cubic bound, are proved with exact computer certificates. After the proof of Theorem 5.5 a worked example shows how one such step becomes a polynomial identity checked by Lean, and the supplement [9] lists each computer-assisted step with its parameter range, its certificate script and the Lean declaration that proves it. The other results have proofs without computer assistance. All theorems, corollaries and propositions, and Lemma 5.4, are formalized in Lean; the declarations are listed in the supplement [9].

What is new, imported and known. New here are the criteria for increasing ratios (Theorem 3.1 with Corollary 3.2, Proposition 3.3, Theorems 4.1, 4.2 and 5.1 with Corollary 5.2, Theorems 5.5 and 5.6 and Corollary 5.7) and the counterexamples of Appendix A. Imported from [5], on which this paper depends, are the finite realization of the arrays [5, Lemma 2.1], the kernel recurrence [5, Lemma 3.2] with the computation in its proof, and the detour inequalities and operator induction [5, Lemma 3.4 and proof of Theorem 3.1], which Section 3 extends; the log-concavity inequality of Lemma 2.1 is also from [5], and its proof is repeated here. Nothing else from [5] is used. Previously known are the constant-ratio case [5], Proposition 5.3, which is Walkup's theorem [12] and is given a self-contained proof in the Newton basis, and the Abel argument of Lemma 5.4, which is that of Wang and Yeh [13].

Table 1 lists the criteria with their hypotheses, the earlier criterion that each extends, and an array or class that it covers and the earlier criteria do not.

Table 1: The criteria of this paper. Each array satisfies (2.1), with profile $d_i = b_i + c_i$ and ratio $\rho(i) = c_i/d_i$. "u LC" means that the Newton coefficients of u are nonnegative and log-concave without internal zeros, and "$b\uparrow$" that b is nondecreasing.

Result	Hypotheses	Extends	Newly covered
[5]	d LC; $\rho \equiv r \in (0,1]$	—	$K_{N,r}$, $G_{N,q,r}$
Thm. 3.1, Cor. 3.2	c LC; Newton coefficients of b are $w_k\kappa_k$ with $w_1 \geq w_2 \geq \cdots \geq 0$	[5] (w constant)	c convex quadratic, $b_i = w_1\kappa_1 i + w_2\kappa_2\binom{i}{2}$, $w_1 > w_2 \geq 0$
Prop. 3.3	b, c quadratic in i; $\kappa_2 \geq 0$, $\kappa_2 + \beta_2 \geq 0$; $b\uparrow$	Cor. 3.2 on quadratic arrays	$d_i = \frac{i}{N}$, $\rho(i) = r + \kappa\frac{i}{N}$, $0 < \kappa \leq \frac{N(1-r)}{2N-1}$, $N \geq 2$
Thm. 4.1	d LC; $\rho(i) = r + \kappa\frac{i}{N}$, $\kappa \geq 0$; $b\uparrow$	Prop. 3.3 (linear d)	$N = 4$, $d_i = \frac{2^i-1}{15}$, $r = \frac{1}{2}$, $\kappa = \frac{2}{5}$
Thm. 4.2	d LC; $\rho = r + tx + \tau\binom{x}{2}$, $t, \tau \geq 0$; $b\uparrow$	Thm. 4.1 ($\tau = 0$)	$d_i = \frac{i}{4}$, $\rho = \frac{1}{2} + \frac{1}{40}\binom{x}{2}$ (c not LC)
Thm. 5.1, Cor. 5.2	$d = \nu_1 i + \nu_2\binom{i}{2}$; c LC (Cor.: ρ LC, any degree); $b\uparrow$	Prop. 3.3 on these profiles	$d_i = \frac{i}{N}$, $\rho = r(1+\varepsilon)^x$
Thm. 5.5	d LC; ρ LC, $\rho(0) > 0$; $b\uparrow$	Thm. 4.1, Cor. 5.2, Thm. 4.2 for $t^2 \geq r\tau$	$\rho = r(1+\varepsilon)^x$ on every LC profile d
Thm. 5.6, Cor. 5.7	d LC; $\rho = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$, $t, \tau, \sigma \geq 0$, $r\sigma \leq t\tau$ (Cor.: $\leq \frac{7}{4}t\tau$); $b\uparrow$	Thm. 4.2 ($\sigma = 0$), Thm. 5.5 for cubic ρ	cubic ρ, $\sigma > 0$, with $t^2 < r\tau$ or $\tau^2 < t\sigma$ (Cor.: also $t\tau < r\sigma \leq \frac{7}{4}t\tau$)

2. Setting

We use the conventions of [5]. An array of real coefficients $(a_i, b_i, c_i)_{0\leq i\leq N}$ satisfies

$$a_0 = 1, \quad b_0 = c_0 = 0, \quad a_N = 0, \quad a_i + b_i + c_i = 1, \qquad a_i > 0 \ (i < N), \qquad c_i > 0 \ (i > 0), \tag{2.1}$$

and $R_{-1} = 0$, $R_0 = 1$ and $xR_i = a_iR_{i+1} + b_iR_i + c_iR_{i-1}$ $(0 \leq i < N)$ define its polynomials, with $R_i(1) = 1$. As in [5] we call such arrays *normalized*: their recurrence matrix has row sums one, but (2.1) allows $b_i < 0$, and we call an array *stochastic* only when all $b_i \geq 0$. By [5, Lemma 2.1] the array has a positive orthogonality measure on $N+1$ real points, which include 1, and on this support $R_mR_n = \sum_k g(m,n;k)R_k$ with $g(m,n;k) = [R_m(J)]_{n,k}$, where J is the tridiagonal recurrence matrix; the array has *nonnegative linearization*, or *property (P)*, if all $g(m,n;k) \geq 0$. We write $d_i = b_i + c_i$ for the profile and, for $i \geq 1$ with $d_i > 0$, $\rho(i) = c_i/d_i$ for the ratio.

The Newton coefficients of a sequence $(u_i)_{0\leq i\leq N}$ are $\Delta^k u(0) = \sum_{i\leq k}(-1)^{k-i}\binom{k}{i}u_i$, so that $u_n = \sum_k \binom{n}{k}\Delta^k u(0)$. A nonnegative, finitely supported sequence $(u_k)_{k\geq 0}$ is *log-concave without internal zeros* if its positive entries form an interval and $u_k^2 \geq u_{k-1}u_{k+1}$ for all

$k \geq 1$. For such a sequence put $f_n(\ell) = \sum_{h\geq 0} \binom{\ell}{h} u_{n+h}$, and, for integers $p, q \geq 0$,

$$W_{p,q}(\ell, a) = \sum_{n\geq p} \binom{n-p+q}{q} \binom{a}{n} f_n(\ell),$$

with $A_j = W_{2j+1,j}$, $B_j = W_{2j+2,j}$ and $C_j = W_{2j+2,j+1}$ when these are formed from the Newton coefficients of the profile.

Lemma 2.1 *([5]). Let $(u_k)_{k\geq 0}$ be nonnegative, finitely supported and log-concave without internal zeros. Then $(u_k + u_{k+1})_{k\geq 0}$ has the same property, hence so does $(f_n(\ell))_{n\geq 0}$ for every $\ell \geq 0$, and*

$$f_i(\ell)\, f_l(\ell+1) \;\geq\; f_0(\ell+1)\, f_{i+l}(\ell) \qquad (i, l \geq 1,\ \ell \geq 0).$$

Proof. For a nonnegative finitely supported sequence u that is log-concave without internal zeros, $v_i = u_i + u_{i+1}$ has the same property, since $v_i^2 - v_{i-1}v_{i+1} = (u_i^2 - u_{i-1}u_{i+1}) + (u_{i+1}^2 - u_i u_{i+2}) + (u_i u_{i+1} - u_{i-1}u_{i+2})$ and the last term is nonnegative by log-concavity. The sequence $(f_n(\ell))_n$ is obtained from $(u_n)_n$ by applying $u \mapsto v$ ℓ times. For such a sequence g, balanced products dominate: $g_i g_l \geq g_0 g_{i+l}$ and $g_i g_{l+1} \geq g_1 g_{i+l}$ for $i, l \geq 1$. Adding these with $g = f(\ell)$ and using $g_l + g_{l+1} = f_l(\ell+1)$ and $g_0 + g_1 = f_0(\ell+1)$ gives the claim. ∎

For the profile, with $u_k = \nu_k$ and $\nu_0 = 0$, one has $f_0(\ell) = d_\ell$, and the inequality reads $f_i(\ell) f_l(\ell+1) \geq d_{\ell+1} f_{i+l}(\ell)$. The paper [5] proves property (P) for constant ratios by expanding the product coefficients in a kernel $\omega_{\ell,a}$ on $\Omega = \{(\ell, a) : \ell, a \geq 0,\ \ell + a \leq N\}$, which satisfies a recurrence in ℓ whose moves have weights formed from the $W_{p,q}$, and by matching each move of odd index with two-step detours through moves of even index [5, Lemmas 3.2 and 3.4]. We extend that argument.

3. Ratios given by weights

The kernel argument also works directly for normalized arrays, with independent diagonal and downward coefficients. Let $a_i + b_i + c_i = 1$ for $0 \leq i \leq N$, with $b_0 = c_0 = 0$, $a_N = 0$, $a_i > 0$ for $i < N$ and $c_i > 0$ for $i \geq 1$. Write

$$\kappa_k = \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} c_i, \qquad \beta_k = \sum_{i=0}^{k} (-1)^{k-i} \binom{k}{i} b_i \qquad (1 \leq k \leq N),$$

with $\kappa_k = \beta_k = 0$ for $k = 0$ and $k > N$, and put $\varphi_n(\ell) = \sum_{h\geq 0} \binom{\ell}{h} \kappa_{n+h}$ and $\psi_n(\ell) = \sum_{h\geq 0} \binom{\ell}{h} \beta_{n+h}$, so that $\varphi_0(\ell) = c_\ell$ and $\psi_0(\ell) = b_\ell$ for $\ell \leq N$.

Theorem 3.1 *(Varying ratio). Let (κ_k) be log-concave without internal zeros and $\beta_k \geq 0$ for all k. Suppose that for $0 \leq \ell \leq N-1$*

$$\psi_1(\ell)\, \varphi_n(\ell+1) \;\geq\; c_{\ell+1}\, \psi_{n+1}(\ell) \quad (n \geq 2), \qquad \varphi_2(\ell)\, \psi_n(\ell+1) \;\geq\; c_{\ell+1}\, \psi_{n+2}(\ell) \quad (n \geq 1).$$

Then the system has nonnegative linearization.

Corollary 3.2. *If (κ_k) is log-concave without internal zeros and $\beta_k = w_k\kappa_k$ for $k \geq 1$ with $w_1 \geq w_2 \geq \cdots \geq 0$, then the system has nonnegative linearization. The ratio $c_i/(b_i + c_i)$ is nondecreasing in $i \geq 1$, and it is constant when w is constant on the support of κ; these are the arrays of [5, Theorem 3.1].*

For example, if c is a convex quadratic in i ($\kappa_1, \kappa_2 > 0$, $\kappa_k = 0$ for $k \geq 3$) and $b_i = w_1\kappa_1 i + w_2\kappa_2\binom{i}{2}$ with $w_1 > w_2 \geq 0$, scaled so that $b_N + c_N = 1$, the ratio $c_i/(b_i + c_i)$ increases strictly. The hypotheses exclude the counterexample recorded in Appendix A, whose diagonal coefficients $b = (0, 1, 1, 1)/91$ have Newton coefficients of both signs. For $N \geq 2$, a strictly increasing linear ratio on a linear profile violates the corollary's assumptions because $\Delta^2 b(0) < 0$; Proposition 3.3 covers it when b is nondecreasing.

Proof of Theorem 3.1. Put $e_i = \prod_{j<i} a_j/c_{j+1}$ and $E_i = e_i R_i$. Then $xE_i = c_{i+1}E_{i+1} + b_iE_i + a_{i-1}E_{i-1}$ on the support. The out-of-range terms are omitted: at the endpoints, $xE_0 = c_1E_1$ and $xE_N = b_NE_N + a_{N-1}E_{N-1}$. The series $F = \sum_i E_i z^i$ satisfies $xF = \mathcal{L}_z F$ with

$$\mathcal{L}_z = z^{-1}c(\theta_z) + b(\theta_z) + z\big(1 - b(\theta_z) - c(\theta_z)\big),$$

where $c(\theta) = \sum_k \kappa_k\binom{\theta}{k}$ and $b(\theta) = \sum_k \beta_k\binom{\theta}{k}$. As in [5, Section 3], $K_k = \sum_{m,n} L^k_{mn} z^m w^n$, with $E_mE_n = \sum_k L^k_{mn}E_k$, satisfies $(\mathcal{L}_z - \mathcal{L}_w)K_k = 0$, and $K_k = \sum \omega_{\ell,a} S^a Z^\ell$ with $\omega_{0,a} = \mathbf{1}_{a=k}$ and $\omega_{\ell,a} = 0$ for $\ell + a > N$. The computation in the proof of [5, Lemma 3.2] gives

$$\frac{(\mathcal{L}_z - \mathcal{L}_w)S^aZ^\ell}{z - w} = S^aZ^\ell + \sum_{p\geq 0}\binom{a}{p}S^{a-p}Z^\ell\Big(\varphi_p(\ell)(\chi_{p-2} - \chi_p) + \psi_p(\ell)(\chi_{p-1} - \chi_p)\Big),$$

with $\chi_{-2} = -Z^{-1}$ and $\chi_{-1} = 0$. Collecting terms as there, with $W_{p,q}$ formed from φ and $W^\psi_{p,q}$ from ψ, gives

$$\begin{aligned} c_{\ell+1}\omega_{\ell+1,a} = a_{\ell+a}\,\omega_{\ell,a} + \sum_{j\geq 0}(-1)^j\Big(&A^\psi_j(\ell - j, a + 2j + 1)\,\omega_{\ell-j,a+2j+1} \\ &+ B_j(\ell - j, a + 2j + 2)\,\omega_{\ell-j,a+2j+2} \\ &+ \big(C_j + C^\psi_j\big)(\ell - j - 1, a + 2j + 2)\,\omega_{\ell-j-1,a+2j+2}\Big), \end{aligned}$$

where $A^\psi_j = W^\psi_{2j+1,j}$, $C^\psi_j = W^\psi_{2j+2,j+1}$, and $B_j = W_{2j+2,j}$, $C_j = W_{2j+2,j+1}$. The straight weight is $1 - c_{\ell+a} - b_{\ell+a} = a_{\ell+a}$, extended polynomially beyond N, and $\beta_0 = 0$. Up to this point no hypothesis of the theorem has been used: the recurrence holds for every normalized array, whatever the signs of its diagonal coefficients. Under the hypotheses, all weights are nonnegative on Ω.

For $j \geq 1$, $0 \leq \ell \leq N - 1$ and $a \geq 0$, the following detour inequalities hold, with weights having a negative second argument interpreted as zero:

$$\begin{aligned} A^\psi_0(\ell, a)B_{j-1}(\ell + 1, a - 1) + B_0(\ell, a)A^\psi_{j-1}(\ell + 1, a - 2) &\geq 4c_{\ell+1}A^\psi_j(\ell, a), \\ B_0(\ell, a)B_{j-1}(\ell + 1, a - 2) &\geq 4c_{\ell+1}B_j(\ell, a), \\ B_0(\ell, a)\big(C_{j-1} + C^\psi_{j-1}\big)(\ell + 1, a - 2) &\geq 4c_{\ell+1}\big(C_j + C^\psi_j\big)(\ell, a). \end{aligned}$$

The proof of [5, Lemma 3.4] applies, with the lower bounds $A_0^{\psi}(\ell,a) \geq a\psi_1(\ell)$ and $B_0(\ell,a) \geq \binom{a}{2}\varphi_2(\ell)$. Lemma 2.1 for φ handles the products $\varphi_2(\ell)\varphi_n(\ell+1)$. The two hypotheses of the theorem handle $\psi_1(\ell)\varphi_n(\ell+1)$ for $n \geq 2$ and $\varphi_2(\ell)\psi_n(\ell+1)$ for $n \geq 1$, which are the indices that occur. The moves C_j and C_j^{ψ} have the same displacement and are treated as one move. With this, the proof of [5, Theorem 3.1] gives $\omega \geq 0$, hence $L_{mn}^k \geq 0$ and $g(m,n;k) = L_{mn}^k e_k/(e_m e_n) \geq 0$. ∎

Proof of Corollary 3.2. Set $w_0 = w_1$; since $\kappa_0 = 0$, this leaves $\beta_0 = 0$. Fix ℓ and write $G_i = \varphi_i(\ell)$, $H_i = \psi_i(\ell)$ and $u_x = \binom{\ell}{x-i}$. For $i \geq 0$,

$$H_iG_{i+1} - H_{i+1}G_i = \sum_{x,y\geq i} u_x u_y \kappa_x \kappa_{y+1}(w_x - w_{y+1}).$$

Terms with $x \leq y$ are nonnegative, and terms with $x = y+1$ vanish. The map $(x,y) \mapsto (y+1, x-1)$ is a bijection from the terms with $x \geq y+2$ onto those with $i+1 \leq x \leq y$; the remaining terms have $x = i$. A term and its image sum to $\kappa_x\kappa_{y+1}(w_{y+1} - w_x)(u_{y+1}u_{x-1} - u_x u_y) \geq 0$, because w is nonincreasing and the zero-extended binomial coefficients are log-concave. Hence $H_{i+1}G_i \leq H_iG_{i+1}$.

Since the support of (G_i) is an interval and $\beta_k = 0$ where $\kappa_k = 0$, chaining gives $H_{n+1}G_1 \leq H_1G_{n+1}$, $H_{n+2}G_n \leq H_nG_{n+2}$ and $H_{n+2}G_{n+1} \leq H_{n+1}G_{n+2}$. Lemma 2.1 with $(i,l) = (1,n)$ and $(2,n)$ reads $G_1(G_n + G_{n+1}) \geq CG_{n+1}$ and $G_2(G_n + G_{n+1}) \geq CG_{n+2}$, where $C = G_0 + G_1 = c_{\ell+1}$ and $G_n + G_{n+1} = \varphi_n(\ell+1)$. Since $G_1 \geq \kappa_1 > 0$, $H_1(G_n + G_{n+1}) \geq CH_1G_{n+1}/G_1 \geq CH_{n+1}$. If $G_{n+2} > 0$, then $G_2(H_n + H_{n+1}) \geq G_2H_{n+2}(G_n + G_{n+1})/G_{n+2} \geq CH_{n+2}$; if $G_{n+2} = 0$, then $H_{n+2} = 0$. These are the two hypotheses of Theorem 3.1. The case $i = 0$ of the first display, with $\beta_0 = \kappa_0 = 0$, reads $b_{\ell+1}c_\ell \leq b_\ell c_{\ell+1}$; for $1 \leq \ell < N$, division by $c_\ell c_{\ell+1} > 0$ shows that b_i/c_i is nonincreasing and $c_i/(b_i + c_i)$ is nondecreasing. If w is constant on the support of κ, then b is proportional to c, and with $d = b + c$ the array is that of [5, Theorem 3.1]. ∎

For arrays that are quadratic in the index, concave diagonal coefficients are allowed.

Proposition 3.3 *(Quadratic arrays).* *Let $b_n = \beta_1 n + \beta_2\binom{n}{2}$ and $c_n = \kappa_1 n + \kappa_2\binom{n}{2}$ $(0 \leq n \leq N)$ define an array as above. If $\kappa_2 \geq 0$, $\kappa_2 + \beta_2 \geq 0$ and $b_0 \leq b_1 \leq \cdots \leq b_N$, then the system has nonnegative linearization.*

Proof. In the recurrence of the proof of Theorem 3.1, which holds for every array, $\varphi_n = \psi_n = 0$ for $n \geq 3$. So all moves with $j \geq 1$ vanish, and the moves with $j = 0$ have the weights $A_0^{\psi}(\ell,a) = \sum_{n\geq 1}\binom{a}{n}\psi_n(\ell) = b_{\ell+a} - b_\ell$, $B_0(\ell,a) = \binom{a}{2}\kappa_2$ and $(C_0 + C_0^{\psi})(\ell,a) = \binom{a}{2}(\kappa_2 + \beta_2)$. These and the straight weights $a_{\ell+a}$ are nonnegative on Ω, so $\omega \geq 0$ by induction on ℓ, and $g(m,n;k) \geq 0$ as in the proof of Theorem 3.1. ∎

With the linear profile $d_n = n/N$ and the ratio $c_n/d_n = r + \kappa n/N$ $(1 \leq n \leq N$, $r > 0$, $\kappa \geq 0)$, one has $\kappa_2 = 2\kappa/N^2$ and $\kappa_2 + \beta_2 = 0$ for $N \geq 2$. For $N = 1$, property (P) follows directly from $R_1^2 = c_1R_0 + b_1R_1$ when $b_1 \geq 0$. So every linearly increasing ratio with a linear profile and nondecreasing b has property (P); b is nondecreasing exactly when $\kappa \leq (1-r)N/(2N-1)$. Exact computations [7] show that each of the three hypotheses

is needed: dropping any one of them gives quadratic arrays with a negative coefficient at $N \le 3$.

4. Linearly and quadratically increasing ratios

For curved profiles, linearly increasing ratios need not satisfy the hypotheses of Theorem 3.1: the Newton coefficients of b of order at least 3 can be negative, and then some moves of even index in the recurrence of its proof have negative weight and no two-step detour. A resummation of the positive moves handles this.

Theorem 4.1 *(Linearly increasing ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, let $0 < r \le 1$ and $\kappa \ge 0$, and put $\rho_i = r + \kappa i/N$, $a_i = 1 - d_i$, $b_i = (1-\rho_i)d_i$ and $c_i = \rho_i d_i$. If b is nondecreasing, then the system has nonnegative linearization. The condition that b be nondecreasing is equivalent to $\kappa\big(N + d_{N-1}/(1-d_{N-1})\big) \le N(1-r)$.*

For the linear profile $d_i = i/N$ the condition reads $\kappa \le \frac{N}{2N-1}(1-r)$, and these are arrays of Proposition 3.3. For a convex profile $d_{N-1} \le (N-1)/N$, so the admitted range of κ contains $[0, \frac{N}{2N-1}(1-r)]$. When $r < 1$, this containment is strict for a nonlinear profile. For example, $N = 4$, $d_i = (2^i - 1)/15$, $r = 1/2$ and $\kappa = 2/5$ are admitted, with $b = (0, 2/75, 3/50, 7/75, 1/10)$, although $(2N-1)\kappa/N = 7/10 > 1 - r$.

Proof. Write $s = 1 - r$ and $t = \kappa/N$, and let $f_n(\ell) = \Delta^n d(\ell)$ and $\nu_i = \Delta^i d(0)$ be as in Section 2, with $\nu_{-1} = \nu_0 = 0$. For $0 \le i < N$ put $R_i = d_i/f_1(i)$; then $f_1(i) \ge \nu_1 > 0$, and log-concavity of $(f_n(i))_{n\ge 0}$ gives $R_{i+1} - R_i = \big(f_1(i)^2 - d_i f_2(i)\big)/\big(f_1(i)f_1(i+1)\big) \ge 0$. Since $b_{i+1} - b_i = f_1(i)\big(s - t(i+1) - tR_i\big)$ and the bracket is nonincreasing in i, b is nondecreasing exactly when its last increment is nonnegative, that is, when $t(N + R_{N-1}) \le s$ with $R_{N-1} = d_{N-1}/(1-d_{N-1})$; this is the stated condition. In particular $tN \le s$. Since $c_i = (r+ti)d_i$ and $b_i = (s-ti)d_i$, the Leibniz rule for differences gives, for $\ell \ge 0$ and $1 \le n \le N - \ell$, which includes every index occurring in the moves on Ω,

$$\varphi_n(\ell) = (r + t(\ell+n))f_n(\ell) + tnf_{n-1}(\ell), \qquad \psi_n(\ell) = h_n(\ell) - v_n(\ell),$$

with $h_n(\ell) = (s - t(\ell+n))f_n(\ell)$ and $v_n(\ell) = tnf_{n-1}(\ell)$; both are nonnegative when $\ell + n \le N$. (Outside this range the formulas need not hold, because the Newton coefficients of b and c are set to zero beyond N.) Extend ν by zero outside $1 \le i \le N$ and put $\widehat{\kappa}_i = p_i\nu_i + q_i\nu_{i-1}$ $(i \ge 1)$, $\widehat{\kappa}_0 = 0$, with $p_i = r + ti$ and $q_i = ti$; the Newton sequence κ of c is the truncation of $\widehat{\kappa}$ at N. Then

$$\begin{aligned}\widehat{\kappa}_i^2 - \widehat{\kappa}_{i-1}\widehat{\kappa}_{i+1} &= p_i^2(\nu_i^2 - \nu_{i-1}\nu_{i+1}) + q_i^2(\nu_{i-1}^2 - \nu_{i-2}\nu_i) \\ &\quad + q_{i-1}p_{i+1}(\nu_i\nu_{i-1} - \nu_{i-2}\nu_{i+1}) + t^2(\nu_{i-1}\nu_{i+1} + \nu_{i-2}\nu_i + 2\nu_i\nu_{i-1}),\end{aligned}$$

and each term is nonnegative by log-concavity of ν. The positive support of $\widehat{\kappa}$ is that of ν when $t = 0$ and extends it by one index when $t > 0$, an interval in both cases. Truncation at N preserves log-concavity and interval support, so κ is log-concave without internal zeros, and Lemma 2.1 applies to f and to φ.

Moves. We use the recurrence of the proof of Theorem 3.1, which holds for every array. For $j \ge 1$ split $A^\psi_j = H_j - V_j$, where $H_j = W^h_{2j+1,j}$ and $V_j = W^v_{2j+1,j}$ are formed from h_n and v_n in place of ψ_n; both are nonnegative on Ω. Keep $A^\psi_0(\ell,a) = b_{\ell+a} - b_\ell$ as one move; it is nonnegative because b is nondecreasing. The weights B_j, formed from φ, and $C_j + C^\psi_j = W^f_{2j+2,j+1}$, formed from f, are nonnegative. Write $C'_j = C_j + C^\psi_j$.

Detours. All detour inequalities below are asserted for source nodes $(\ell,a) \in \Omega$; in the mixed products the occurring indices satisfy $n \le a-2$, hence $\ell+n+2 \le N$. Since $\varphi_2(\ell) \ge (r+t(\ell+1))f_2(\ell)$ and $c_{\ell+1} = (r+t(\ell+1))d_{\ell+1}$, Lemma 2.1 for f gives

$$\varphi_2(\ell)f_n(\ell+1) \ge c_{\ell+1}f_{n+2}(\ell), \qquad \varphi_2(\ell)h_n(\ell+1) \ge c_{\ell+1}h_{n+2}(\ell),$$
$$\varphi_2(\ell)v_n(\ell+1) \ge \tfrac{n}{n+2}\, c_{\ell+1}v_{n+2}(\ell) \quad (n \ge 2),$$

using $s - t(\ell+n+1) \ge s - t(\ell+n+2) \ge 0$ for the second and Lemma 2.1 with indices $(2, n-1)$ for the third. As in the proof of [5, Lemma 3.4], with $B_0(\ell,a) \ge \binom{a}{2}\varphi_2(\ell)$, for $0 \le \ell \le N-1$ and second factors at $(\ell+1, a-2)$,

$$B_0B_{j-1} \ge 4c_{\ell+1}B_j \ (j \ge 1), \qquad B_0H_{j-1} \ge 6c_{\ell+1}H_j \ (j \ge 2),$$
$$B_0V_{j-1} \ge 6c_{\ell+1}V_j \ (j \ge 2), \qquad B_0C'_{j-1} \ge 6c_{\ell+1}C'_j \ (j \ge 1).$$

The first uses Lemma 2.1 for φ. The coefficient ratios are, on $m \ge 2j+2$, $2j+1$, $2j+1$ and $2j+2$ respectively,

$$\frac{jm(m-1)}{2(m-j-2)}, \qquad \frac{jm(m-1)}{2(m-j-1)}, \qquad \frac{j(m-1)(m-2)}{2(m-j-1)}, \qquad \frac{(j+1)m(m-1)}{2(m-j-1)};$$

they increase with j and are at least 5, 10, 6 and 6. Finally,

$$A^\psi_0(\ell,a)B_0(\ell+1,a-1) + B_0(\ell,a)A^\psi_0(\ell+1,a-2) \ge c_{\ell+1}\big(4H_1(\ell,a) - 2V_1(\ell,a)\big). \quad (4.1)$$

Only $a \ge 3$ needs proof. Put $u_n = f_n(\ell)$, $y_n = \binom{a}{n}u_n$, $D = u_0 + u_1 = d_{\ell+1}$, $\alpha = s - t(\ell+a)$,

$$X = \sum_{n\ge1} y_n, \qquad Y = \sum_{n\ge2} y_n, \qquad J = \sum_{n=2}^{a-1} \tbinom{a-1}{n}u_n,$$
$$X' = \sum_{j\ge1} \tbinom{a-2}{j}(u_j + u_{j+1}), \qquad Y' = \sum_{j\ge2} \tbinom{a-1}{j}(u_j + u_{j+1}),$$

so that $X = au_1 + Y$, $X' = (a-2)u_1 + J$ and $Y' = Y - (a-1)u_2$. Then $A^\psi_0(\ell,a) = \alpha X - tau_0$ and $A^\psi_0(\ell+1,a-2) = (\alpha+t)X' - t(a-2)D$; both are nonnegative, and the two weights B_0 are at least $(r+t(\ell+1))Y'$ and $(r+t(\ell+1))Y$. With $H_* = \sum_{n\ge3}(n-2)y_n$ and $T = \sum_{n\ge2}(n-3)(a-n)y_n$ one has $H_1 = \sum_{n\ge3}(n-2)(\alpha+t(a-n))y_n$ and, after a shift of index, $V_1 = t\sum_{n\ge2}(n-1)(a-n)y_n$, so $4H_1 - 2V_1 = 4\alpha H_* + 2tT$. Since $c_{\ell+1} = (r+t(\ell+1))D$, it suffices to show

$$(\alpha X - tau_0)Y' + Y\big((\alpha+t)X' - t(a-2)D\big) \ge D(4\alpha H_* + 2tT). \quad (4.2)$$

The coefficient of α in the difference of the two sides is $K = XY' + YX' - 4DH_*$. By Lemma 2.1, $u_i(u_j + u_{j+1}) \ge Du_{i+j}$ for $i,j \ge 1$, so $XY' \ge au_1Y' \ge D\sum_{n\ge3} ny_n$ and

$YX' \ge \binom{a}{2}u_2X' \ge D\sum_{n\ge3}\binom{n}{2}y_n$; hence $K \ge \frac{D}{2}\sum_{n\ge3}\big((n-3)(n-4)+4\big)y_n \ge 0$, and (4.2) holds when $t=0$. Let $t>0$. Monotonicity of b at $\ell+a-1$ gives $\alpha \ge tR_{\ell+a-1} \ge tR_\ell = tu_0/u_1$, so it suffices to prove (4.2) at $\alpha = tu_0/u_1$. Both sides are homogeneous of degree two in (u_n); take $u_1=1$ and write $z=u_0$ and $q=u_2$. If $q=0$, then $u_n=0$ for $n\ge2$ and both sides vanish. Otherwise $0\le z\le 1/q$ by log-concavity. At $\alpha=tz$ the two weights A_0^ψ are tzY and $t(z+1)J$, and (4.2) divided by t reads $F(z)\ge0$, where

$$\begin{aligned} F(z) &= zYY' + (z+1)YJ - (z+1)(4zH_* + 2T) \\ &= (1-qz)F(0) + qzF(1/q) + 4H_*z(1/q - z), \end{aligned}$$

the second form holding because F is quadratic with leading coefficient $-4H_*$. It remains to show $F(0)\ge0$ and $F(1/q)\ge0$. Put $L_3=\frac{3}{2}$, $L_n=\frac{n(n+2)}{6}$ for $n\ge4$, and $E_n=\frac{2(n^2-5)}{n+1}$. In YJ keep, for each n, the product with indices $(2,2)$ if $n=3$, and the products with indices $(2,n-1)$ and $(3,n-2)$ if $n\ge4$; by log-concavity with $u_1=1$, $u_iu_{n+1-i}\ge u_n$, and $\binom{a-1}{j}\ge\binom{a-i}{j}$, so $YJ\ge\sum_{n=3}^{a-1}L_n(a-n)y_n$. In Y keep the term $\binom{a}{2}q$; as $q(u_{n-1}+u_n)\ge(q+1)u_n$ for $n\ge3$, $YY'\ge(q+1)\sum_{n=3}^{a}\frac{n(a-1)}{2}y_n$. Since $L_n\ge2(n-3)$,

$$F(0) = YJ - 2T \ge 2(a-2)y_2 + \sum_{n=3}^{a-1}\big(L_n - 2(n-3)\big)(a-n)y_n \ge 0.$$

Using $u_n\le qu_{n-1}$ and a shift of index, $\frac{4}{q}H_* + 2T \le \sum_{n=2}^{a-1}E_n(a-n)y_n$, where $E_2<0$. As $L_n+\frac{n}{2}\ge E_n$ for $n\ge3$,

$$\frac{F(1/q)}{1+1/q} = YJ + \frac{YY'}{q+1} - \frac{4}{q}H_* - 2T \ge \sum_{n=3}^{a-1}\Big(L_n + \frac{n}{2} - E_n\Big)(a-n)y_n \ge 0.$$

The numerical facts used are $L_3=\frac{3}{2}\ge0$, $L_3+\frac{3}{2}=3\ge E_3=2$, and, for $n\ge4$, $6(L_n-2(n-3)) = (n-5)^2+11$ and $6(n+1)(L_n+\frac{n}{2}-E_n) = n^3-6n^2+5n+60>0$.

Resummation. Regard the moves as operators on functions on Ω, each divided by c at the destination level; all strictly raise ℓ, and inequalities between operators are entrywise. Let P and Q be the sums of the B_j with j even and odd, $C=C_+-C_-$ with $C_\pm$ the sums of the C_j' with j even and odd, H_+ and H_- the sums of the H_j with $j\ge2$ even and $j\ge3$ odd, V_+ and V_- the sums of the V_j with j odd and $j\ge2$ even, A_0 the operator of A_0^ψ, and U the straight moves. The recurrence reads $\omega=\beta+(U+P-Q+C+A)\omega$ with $A=A_0+H_+-H_--H_1+V_+-V_-$ and $\beta=\omega|_{\ell=0}\ge0$. The inequalities above give

$$P^2\ge4Q,\quad H_+P\ge6H_-,\quad V_+P\ge6V_-,\quad C_+P\ge6C_-,\quad A_0P+PA_0+2V_1\ge4H_1.$$

Let $G=(I-P/2)^{-1}=\sum_{m=0}^{N}(P/2)^m\ge0$. Then $PG=GP=2(G-I)$ and $(I-P/6)G=\frac{2}{3}G+\frac{1}{3}I$, so $G(H_+-H_-)G\ge GH_+(I-P/6)G\ge0$, $GCG\ge0$ and $G(V_+-V_-)G\ge\frac{2}{3}GV_1G$. Hence

$$\begin{aligned} GAG &\ge G\big(A_0 - H_1 + \tfrac{2}{3}V_1\big)G \ge G\big(A_0 - \tfrac{1}{4}(A_0P+PA_0) + \tfrac{1}{6}V_1\big)G \\ &= \tfrac{1}{2}(GA_0 + A_0G) + \tfrac{1}{6}GV_1G \ge 0. \end{aligned}$$

With $R = U + P^2/4 - Q \geq 0$, the operator $M = G(R + C + A)G$ is nonnegative and strictly raises ℓ. Since $(I - P/2)^2\omega = \beta + (R + C + A)\omega$, the function $v = (I - P/2)\omega$ satisfies $v = G\beta + Mv$. Therefore $\omega = G\sum_{q=0}^{N} M^q G\beta \geq 0$, and $g(m, n; k) \geq 0$ as in the proof of Theorem 3.1. ∎

Theorem 4.2 *(Convex quadratic ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, let $\rho(x) = r + tx + \tau\binom{x}{2}$ with $r > 0$ and $t, \tau \geq 0$, and put $a_i = 1 - d_i$, $b_i = (1 - \rho(i))d_i$ and $c_i = \rho(i)d_i$. If b is nondecreasing, then the system has nonnegative linearization.*

This contains Theorem 4.1 ($\tau = 0$). As in the proof of that theorem, $b_{i+1} - b_i = f_1(i)\big(1 - \rho(i+1) - (t + \tau i)R_i\big)$ with a nonincreasing bracket, so b is nondecreasing exactly when $t(N + R) + \tau\big(\binom{N}{2} + (N-1)R\big) \leq 1 - r$ with $R = d_{N-1}/(1 - d_{N-1})$. The Newton coefficients of c need not be log-concave: for $d_i = i/4$ and $\rho(x) = \frac{1}{2} + \binom{x}{2}/40$ they are $(1/8, 1/80, 3/160, 0)$. The proof below replaces their log-concavity by a weighted detour inequality.

Proof. Write $f_n(\ell) = \Delta^n d(\ell)$, $\Delta\rho(x) = t + \tau x$ and $R_i = d_i/f_1(i)$, which is nondecreasing as in the proof of Theorem 4.1. Since $b_{i+1} - b_i = f_1(i)\big(1 - \rho(i+1) - \Delta\rho(i)R_i\big)$ and b is nondecreasing, $1 - \rho(m+1) \geq \Delta\rho(m)R_m$ for $0 \leq m < N$, and $\rho(i) \leq 1$ for $i \leq N$. For $\ell + n \leq N$ the Leibniz rule gives, all at ℓ,

$$\varphi_n = \rho(\ell+n)f_n + n\Delta\rho(\ell+n-1)f_{n-1} + \binom{n}{2}\tau f_{n-2}, \qquad \psi_n = h_n - v_n - w_n,$$

with $h_n = (1 - \rho(\ell+n))f_n$, $v_n = n\Delta\rho(\ell+n-1)f_{n-1}$ and $w_n = \binom{n}{2}\tau f_{n-2}$; here φ_n, h_n, v_n and w_n are nonnegative, while ψ_n need not be.

Moves. In the recurrence of the proof of Theorem 3.1, keep $A_0^{\psi} = b_{\ell+a} - b_\ell \geq 0$ and split $A_j^{\psi} = H_j - V_j - W_j$ for $j \geq 1$, where H_j, V_j and W_j are $W_{2j+1,j}$ formed from h_n, v_n and w_n; the weights B_j, formed from φ, and $C'_j = C_j + C_j^{\psi} = W^f_{2j+2,j+1}$ are nonnegative. With the signs $(-1)^j$, the moves W_j behave like the V_j.

Detours of H, V, W and C'. All detours are asserted for sources $(\ell, a) \in \Omega$, with second factors at $(\ell + 1, a - 2)$, marked by a prime. Since $\varphi_2(\ell) \geq \rho(\ell+1)f_2(\ell)$ and $c_{\ell+1} = \rho(\ell+1)d_{\ell+1}$, Lemma 2.1 for f gives, for $n \leq a - 2$,

$$\varphi_2 f'_n \geq c_{\ell+1}f_{n+2}, \qquad \varphi_2 h'_n \geq c_{\ell+1}h_{n+2},$$
$$(n+1)(n+2)\varphi_2 v'_n \geq n^2 c_{\ell+1}v_{n+2}, \qquad (n+1)(n+2)\varphi_2 w'_n \geq n(n-1)c_{\ell+1}w_{n+2},$$

the last two for $n \geq 3$. The second uses $1 - \rho(\ell+n+1) \geq 1 - \rho(\ell+n+2) \geq 0$; the third uses Lemma 2.1 with indices $(2, n-1)$ and $(n+1)\Delta\rho(\ell+n) - n\Delta\rho(\ell+n+1) = t + \tau\ell \geq 0$; the fourth uses indices $(2, n-2)$. As in the proof of [5, Lemma 3.4], with $B_0(\ell, a) \geq \binom{a}{2}\varphi_2(\ell)$ and $\binom{a}{2}\binom{a-2}{m-2} = \binom{a}{m}\binom{m}{2}$,

$$B_0 H'_{j-1} \geq 10c_{\ell+1}H_j, \qquad B_0 V'_{j-1} \geq \tfrac{9}{2}c_{\ell+1}V_j, \qquad B_0 W'_{j-1} \geq 3c_{\ell+1}W_j \quad (j \geq 2),$$
$$B_0\, C'_{j-1}(\ell+1, a-2) \geq 6c_{\ell+1}C'_j \quad (j \geq 1).$$

The coefficient ratios are $\frac{jm(m-1)}{2(m-j-1)}$, $\frac{j(m-2)^2}{2(m-j-1)}$ and $\frac{j(m-2)(m-3)}{2(m-j-1)}$ on $m \geq 2j+1$, and $\frac{(j+1)m(m-1)}{2(m-j-1)}$ on $m \geq 2j+2$; they increase with j, and at the smallest j the bounds are $(m-5)(m-6) \geq 0$, $(m-5)(2m-7) \geq 0$, $m \geq 5$ and $(m-3)(m-4) \geq 0$.

Detour of B. We show, for $n \geq 2$ and $\ell + n + 2 \leq N$,

$$(n+1)(n+2)\,\varphi_2(\ell)\varphi_n(\ell+1) \;\geq\; 8(n-1)\,c_{\ell+1}\varphi_{n+2}(\ell). \tag{4.3}$$

Granting this, $B_0(\ell,a) \geq \binom{a}{2}\varphi_2(\ell)$, (4.3) with $n = m-2$, $\binom{a}{2}\binom{a-2}{m-2} = \binom{a}{m}\binom{m}{2}$ and $\binom{m}{2}\frac{8(m-3)}{m(m-1)} = 4(m-3)$ give

$$\begin{aligned} B_0(\ell,a)B_{j-1}(\ell+1,a-2) \;&\geq\; 4c_{\ell+1}\sum_{m\geq 2j+2}(m-3)\binom{m-j-3}{j-1}\binom{a}{m}\varphi_m(\ell) \\ &\geq\; 4c_{\ell+1}B_j(\ell,a) \end{aligned}$$

for $j \geq 1$, because $(m-3)\binom{m-j-3}{j-1} - \binom{m-j-2}{j} = \binom{m-j-3}{j-1}\frac{(j-1)(m-2)}{j} \geq 0$. (With constant 1 in place of $\frac{(n+1)(n+2)}{8(n-1)}$, (4.3) can fail: for $N = 20$, $d_i = (2^i-1)/(2^{20}-1)$, $r = \frac{1}{2}$, $t = 0$, $\tau = 10^{-3}$ one gets $\varphi_2(16)\varphi_2(17) < c_{17}\varphi_4(16)$.)

To prove (4.3), put $u_k = f_k(\ell)$, so that $f_k(\ell+1) = u_k + u_{k+1}$, and $p = \rho(\ell+1)$, $K_n = (n+1)(n+2)$, $L_n = 8(n-1)$. Both sides are homogeneous of degree two in u; take $u_1 = 1$ and write $z = u_0$, $q = u_2$. By convexity and log-concavity, $0 \leq z \leq \ell$ and $zq \leq 1$. If $q = 0$ or $u_n = 0$, then $\varphi_{n+2}(\ell) = 0$. Otherwise put $y = u_{n+1}/u_n \in [0,q]$. Log-concavity gives $u_{n+2} \leq y^2 u_n$, and for $n \geq 3$ also $u_{n-1} \geq u_n/q$ and $u_{n-2} \geq u_n/q^2$. Hence $\varphi_n(\ell+1) \geq u_n\mathcal{L}(y)$ and $\varphi_{n+2}(\ell) \leq u_n\mathcal{T}(y)$, where

$$\mathcal{T}(y) \;=\; \rho(\ell+n+2)y^2 + (n+2)\Delta\rho(\ell+n+1)y + \binom{n+2}{2}\tau,$$

$\mathcal{L}(y) = \rho(\ell+n+1)(1+y) + n\Delta\rho(\ell+n)(1+q^{-1}) + \binom{n}{2}\tau(q^{-1}+q^{-2})$ for $n \geq 3$, and $\mathcal{L}(y) = \rho(\ell+3)(1+y) + 2\Delta\rho(\ell+2)(1+q^{-1}) + \tau(z+1)q^{-1}$ for $n = 2$ (exact). In both cases $\mathcal{L}(y) = A + By$ with $A \geq B \geq 0$, so $\mathcal{L}(0) \geq \mathcal{L}(q)/(1+q)$; and $\mathcal{T}(q) \geq (1+q)\mathcal{T}(0)$ because $(n+2)\Delta\rho(\ell+n+1) \geq (n+2)(n+1)\tau$. The function $\mathcal{F}(y) = K_n\varphi_2(\ell)\mathcal{L}(y) - L_n p(z+1)\mathcal{T}(y)$ is concave with $\mathcal{F}(0) \geq \mathcal{F}(q)/(1+q)$, so it suffices that $\mathcal{F}(q) \geq 0$, which is (4.3) for the configuration $u_0 = z$, $u_k = q^{k-1}$ $(k \geq 1)$.

Next we reduce to $z = \min(\ell, 1/q)$. For $n \geq 3$, only $\varphi_2(\ell) = A_2 + \tau z$ and $c_{\ell+1} = p(z+1)$ depend on z, where $A_2 = \rho(\ell+2)q + 2\Delta\rho(\ell+1) \geq \tau$, so $\varphi_2(\ell)/(z+1)$ is non-increasing in z. For $n = 2$, write also $\varphi_2(\ell+1) = G_2 + \tau(z+1)$ with $G_2 = \rho(\ell+3)q(1+q) + 2\Delta\rho(\ell+2)(1+q)$. For $z \leq Z \leq \ell$ the quotient $\varphi_2(\ell)\varphi_2(\ell+1)/(z+1)$ at z minus its value at Z equals $(Z-z)\big(G_2(A_2-\tau)/((z+1)(Z+1)) - \tau^2\big) \geq 0$, since $G_2(A_2-\tau) \geq 2(\ell+2)(2\ell+1)\tau^2 \geq (\ell+1)^2\tau^2$.

For the remaining configurations we use the following polynomial inequality. For an integer $n \geq 2$ and reals $h \geq 1$, $p, \tau \geq 0$ and $s \geq (h-1)\tau$, put $A = p + (2h-1)s + h^2\tau$, $B = p + nh(s+\tau) + \binom{n}{2}h^2\tau$ and $C = p + ((n+2)h-1)s + \binom{n+2}{2}h^2\tau$. Then

$$K_n AB - L_n pC \;\geq\; 0. \tag{4.4}$$

Indeed, $K_nAB - L_npC$ is nondecreasing in s (its ps-coefficient is $a_n((n+2)h-1)$ with $a_n = n^2-5n+10 > 0$, and its other s-dependent coefficients are nonnegative), and at $s = (h-1)\tau$ it equals $a_np^2 + C_n(h)p\tau + D_n(h)\tau^2$, where $C_n(h) = \frac{1}{2}\big(h^2U_n + h(10n^2+14n-60) + 2a_n\big)$, $U_n = n^4-4n^3-21n^2+12n+60$ and $D_n(h) = \frac{1}{2}h^2n(n+1)^2(n+2)(3h^2-3h+1)$. For $n \ge 7$ all three coefficients are nonnegative, since $U_n = k^4+24k^3+189k^2+502k+144$ with $k = n-7$. For $2 \le n \le 6$, $4a_nD_n - C_n^2$ is a polynomial in $v = h-1$ with positive coefficients:

n	v^0	v^1	v^2	v^3	v^4
2	576	2880	5616	4992	1664
3	1520	6240	9744	7200	2160
4	6300	27000	42660	30600	8604
5	25200	126000	229500	184800	56000
6	62720	376320	774144	677376	216576

so $a_np^2 + C_np\tau + D_n\tau^2 = a_n\big(p + \frac{C_n\tau}{2a_n}\big)^2 + \frac{4a_nD_n - C_n^2}{4a_n}\tau^2 \ge 0$.

If $1/q \le \ell$, take $z = 1/q$, $h = 1+1/q$ and $s = t + \tau\ell \ge (h-1)\tau$. For the configuration above, $\mathcal{F}(q)$ divided by $q(1+q)$ equals $K_nAB - L_npC$, which is nonnegative by (4.4).

If $\ell < 1/q$, take $z = \ell$, and write $w = 1/q - \ell > 0$, $h_0 = \ell+1$, $h = h_0 + w$ and $s = t + \tau\ell$. With $A(w) = p + (2h-1)s + \tau\big((\ell+2)h - \ell - 1\big)$ and $B(w)$, $C(w)$ as above (with this h), $\mathcal{F}(q)$ divided by q^2 equals $\mathcal{E}_n(w) = K_n(h_0+w)A(w)B(w) - L_nh_0pC(w)$ for $n \ge 3$, and $\mathcal{E}_2(w) - 12A(w)\tau w(\ell+w)^2$ for $n = 2$. At $w = 0$ both equal $h_0(K_nAB - L_npC)$ with $h = h_0$, which is nonnegative by (4.4). Expanding in w, with $\bar{A} = A(0) - p$ and $\bar{B} = B(0) - p$, every coefficient of w, w^2, w^3 and w^4 is a sum of nonnegative terms: for $n \ge 3$,

$$\begin{aligned}
[w] &= K_np^2 + p(a_1t + b_1\tau) + K_n\big(\bar{A}\bar{B} + h_0(\eta\bar{B} + \bar{A}\theta)\big),\\
[w^2] &= p(a_2t + b_2\tau) + K_n\big(\eta\bar{B} + \bar{A}\theta + h_0(\eta\theta + \bar{A}\epsilon)\big),\\
[w^3] &= K_n\big(\eta\theta + A(0)\epsilon + h_0\eta\epsilon\big), \qquad [w^4] = K_n\eta\epsilon,
\end{aligned}$$

where

$$\begin{aligned}
\eta &= 2s + (\ell+2)\tau, \qquad \theta = n(s+\tau) + n(n-1)\tau h_0, \qquad \epsilon = \tbinom{n}{2}\tau,\\
a_1 &= (n+2)\big(2\ell(n^2-n+6) + 2n^2 - 3n + 11\big), \qquad a_2 = K_n(n+2),\\
b_1 &= \tfrac{n+2}{2}(\ell^2P_n + \ell Q_n + R_n), \qquad b_2 = \tfrac{K_n}{2}\big(3n(n-3)(\ell+1) + 14\ell + 12\big),
\end{aligned}$$

with $P_n = 3k^3+15k^2+6k+8$, $Q_n = 6k^3+30k^2+20k+16$ and $R_n = 3k^3+15k^2+16k+16$ for $k = n-3$; for $n = 2$ the coefficients of w, w^2 and w^3 are

$$\begin{aligned}
&12p^2 + 4(16\ell+13)pt + 8(\ell+1)(2\ell+3)p\tau + 48(\ell+1)(3\ell+2)t^2\\
&\qquad + 12(36\ell^3+92\ell^2+76\ell+21)t\tau + 12(27\ell^4+93\ell^3+114\ell^2+62\ell+13)\tau^2,\\
&48pt + 12(2\ell+3)p\tau + 24(6\ell+5)t^2\\
&\qquad + 12(36\ell^2+67\ell+29)t\tau + 12(27\ell^3+78\ell^2+67\ell+19)\tau^2,\\
&12\big(2s + (\ell+2)\tau\big)\big(2t + (3\ell+5)\tau\big).
\end{aligned}$$

This proves (4.3).

Key inequality. We show, for $(\ell,a)\in\Omega$ with $a\ge 3$,

$$A_0^{\psi}(\ell,a)B_0(\ell+1,a-1)+B_0(\ell,a)A_0^{\psi}(\ell+1,a-2)\ \ge\ c_{\ell+1}\big(4H_1-2V_1-W_1\big)(\ell,a). \quad (4.5)$$

Use the notation of the proof of Theorem 4.1 ($u_n=f_n(\ell)$, y_n, D, X, Y, X', Y', J, H_*, T), and put $\alpha=1-\rho(\ell+a)$, $s=\Delta\rho(\ell+a-1)$, $p=\rho(\ell+1)$ and $Z=\sum_{n\ge2}\binom{a-n}{2}y_n$. Then $\alpha\ge sR_{\ell+a-1}\ge su_0/u_1$ and $s\ge\tau(a-1)$. Backward differences give

$$\begin{aligned}A_0^{\psi}(\ell,a)\ &=\ \alpha X-sau_0+\tau\tbinom{a}{2}u_0,\\ A_0^{\psi}(\ell+1,a-2)\ &=\ (\alpha+s)X'-s(a-2)D+\tau\tbinom{a-1}{2}D,\end{aligned}$$

both nonnegative, and $B_0(\ell,a)\ge pY$, $B_0(\ell+1,a-1)\ge pY'$, $c_{\ell+1}=pD$. Since $1-\rho(\ell+n)=\alpha+(a-n)s-\tau\binom{a-n}{2}$, reindexing gives $4H_1-2V_1=4\alpha H_*+2sT+4\tau Z$ and $W_1=\tau\sum_{n\ge1}n\binom{a-n}{2}y_n$. The argument of the proof of Theorem 4.1 with s in place of t, keeping the surplus of the two endpoints ($F(0)\ge 2(a-2)y_2$ and $F(1/q)\ge\frac23(1+1/q)(a-2)y_2$, from the term $n=2$ and $E_2=-\frac23$), shows

$$(\alpha X-sau_0)Y'+Y\big((\alpha+s)X'-s(a-2)D\big)-D(4\alpha H_*+2sT)\ \ge\ \tfrac23 sD(a-2)y_2$$

for $\alpha u_1\ge su_0$; the right side of this bound is linear in $z=u_0/u_1$ and is subtracted before the interpolation. With $s\ge\tau(a-1)$, the difference of the two sides of (4.5) divided by p is therefore at least

$$\begin{aligned}\tau\Big(\tbinom{a}{2}u_0Y'+D\Big[\tbinom{a-1}{2}y_1+\tfrac{(a-2)(a+11)}{6}y_2+(2a-5)y_3\\ +\sum_{n\ge4}\Big(\tbinom{a-1}{2}+(n-4)\tbinom{a-n}{2}\Big)y_n\Big]\Big)\ \ge\ 0.\end{aligned}$$

Resummation. With $W_\pm$ the sums of the W_j with j odd and $j\ge2$ even, the recurrence reads as in the proof of Theorem 4.1 with $A=A_0-H_1+(H_+-H_-)+(V_+-V_-)+(W_+-W_-)$, and the inequalities above give $P^2\ge4Q$, $H_+P\ge10H_-$, $V_+P\ge\frac92V_-$, $W_+P\ge3W_-$, $C_+P\ge6C_-$ and $A_0P+PA_0+2V_1+W_1\ge4H_1$. Since $(I-P/\mu)G=(1-\frac2\mu)G+\frac2\mu I$, one has $G(H_+-H_-)G\ge0$, $GCG\ge0$, $G(V_+-V_-)G\ge\frac59GV_1G$ and $G(W_+-W_-)G\ge\frac13GW_1G$, hence

$$\begin{aligned}GAG\ &\ge\ G\big(A_0-\tfrac14(A_0P+PA_0)+\tfrac{1}{18}V_1+\tfrac{1}{12}W_1\big)G\\ &=\ \tfrac12(GA_0+A_0G)+\tfrac{1}{18}GV_1G+\tfrac{1}{12}GW_1G\ \ge\ 0.\end{aligned}$$

With $U+P^2/4-Q\ge0$ this gives $M\ge0$, and (P) follows as before. ∎

5. Ratios of higher degree

For profiles of Newton degree at most two the ratio can have any degree, provided the Newton coefficients of c are nonnegative and log-concave without internal zeros; no key inequality is needed.

Theorem 5.1 *(Low-degree profiles).* *Let $d_i = \nu_1 i + \nu_2 \binom{i}{2}$ with $\nu_1 > 0$, $\nu_2 \geq 0$ and $d_N = 1$, let $c_0 = 0$, and suppose that the Newton coefficients $\kappa_k = \Delta^k c(0)$, $1 \leq k \leq N$, are log-concave without internal zeros. Put $a_i = 1 - d_i$ and $b_i = d_i - c_i$. If b is nondecreasing, then the system has nonnegative linearization.*

Proof. We use the recurrence of the proof of Theorem 3.1, which holds for every array, with $\varphi_n(\ell) = \Delta^n c(\ell)$, $\psi_n(\ell) = \Delta^n b(\ell)$ and $f_n(\ell) = \Delta^n d(\ell)$ for the polynomial extensions. Since $f_n = 0$ for $n \geq 3$, every move A_j^{ψ} with $j \geq 1$, which involves only $n \geq 2j+1 \geq 3$, equals $-\widetilde{B}_j$ with $\widetilde{B}_j = W_{2j+1,j}$ formed from φ; the moves $C_j + C_j^{\psi} = W_{2j+2,j+1}^{f}$ vanish for $j \geq 1$, and $C_0 + C_0^{\psi} = \binom{a}{2}\nu_2 \geq 0$. The weight $A_0^{\psi}(\ell, a) = b_{\ell+a} - b_\ell$ is nonnegative on Ω, as are the straight weights and all B_j and $\widetilde{B}_j$, since $(\varphi_n(\ell))_n$ is obtained from (κ_k) by the shifts of Lemma 2.1 and is therefore nonnegative and log-concave without internal zeros. Lemma 2.1 for φ gives, as in [5, Lemma 3.4], $B_0(\ell, a)B_{j-1}(\ell+1, a-2) \geq 4c_{\ell+1}B_j(\ell, a)$ for $j \geq 1$, and with $B_0(\ell, a) \geq \binom{a}{2}\varphi_2(\ell)$ also $B_0(\ell, a)\widetilde{B}_{j-1}(\ell+1, a-2) \geq 4c_{\ell+1}\widetilde{B}_j(\ell, a)$ for $j \geq 2$, because the coefficient quotients $jm(m-1)/(2(m-j-1))$ on $m \geq 2j+1$ are at least 10. In the resummation of the proof of Theorem 4.1 take $H = 0$ and $V_j = \widetilde{B}_j$; the key inequality is then trivial, and $M \geq 0$ gives (P). ∎

Corollary 5.2. *In Theorem 5.1 let $c_i = \rho(i)d_i$ with $\rho(x) = \sum_k \rho_k \binom{x}{k}$, where $(\rho_k)_{k\geq 0}$ is nonnegative and log-concave without internal zeros and $\rho_0 > 0$. Then the Newton coefficients of c are log-concave without internal zeros. Hence every such ratio, of any degree, with nondecreasing b gives nonnegative linearization on these profiles.*

Proof. Put $s_k = \rho_k + \rho_{k-1}$ ($\rho_{-1} = 0$), which is log-concave without internal zeros, and $h = \nu_2/2$. The Leibniz rule gives $\kappa_k = k e_k$ for $k \geq 1$, with $e_k = (\nu_1 + h(k-1))s_k + h(k-1)s_{k-1}$. Write $\alpha = \nu_1 + h(k-1)$ and $\beta = h(k-1)$; then

$$\begin{aligned} e_k^2 - e_{k-1}e_{k+1} &= \alpha^2(s_k^2 - s_{k-1}s_{k+1}) + \beta^2(s_{k-1}^2 - s_{k-2}s_k) \\ &\quad + (\beta - h)(\alpha + h)(s_{k-1}s_k - s_{k-2}s_{k+1}) \\ &\quad + h^2(s_{k-1}s_{k+1} + s_{k-2}s_k + 2s_{k-1}s_k), \end{aligned}$$

and every term is nonnegative for $k \geq 2$ by log-concavity of s. So $\kappa_k^2 \geq k^2 e_{k-1}e_{k+1} \geq \kappa_{k-1}\kappa_{k+1}$. If $\kappa_k = 0$ with $k \geq 1$, then $s_k = 0$, so $\rho_k = 0$; as $\rho_0 > 0$ and ρ has no internal zeros, $\rho_m = 0$ for $m \geq k$, and $\kappa_{k+1} = 0$. ∎

For example, on the linear profile $d_i = i/N$ the exponential ratios $\rho(x) = r(1+\varepsilon)^x$ with $r > 0$ and $\varepsilon \geq 0$, whose Newton coefficients $r\varepsilon^k$ are geometric, give nonnegative linearization whenever b is nondecreasing. Nonnegativity of the ratio's Newton coefficients alone does not suffice: the cubic counterexample in Appendix A has the linear profile $d_i = i/4$ and the ratio with Newton coefficients $(\frac{19}{20}, 0, 0, \frac{1}{320})$, which are nonnegative and satisfy the log-concavity inequalities but have internal zeros. Theorem 4.2 and Corollary 5.2 are not comparable: the former admits convex quadratic ratios with $t^2 < r\tau$ on every log-concave Newton profile, the latter ratios of every degree on profiles of Newton degree at most two.

Products of Newton sequences. The proof of Corollary 5.2 used only that the Newton coefficients of $c = \rho d$ are log-concave. This holds for every profile.

Proposition 5.3 *(Products).* *Let $(\rho_i)_{i\ge 0}$ and $(\nu_j)_{j\ge 0}$ be nonnegative and log-concave without internal zeros, extended by zero at negative indices, and let $f(x) = \sum_i \rho_i \binom{x}{i}$ and $g(x) = \sum_j \nu_j \binom{x}{j}$ for $x \in \mathbb{Z}_{\ge 0}$, where the sums are finite. Then the Newton coefficients $\kappa_k = \Delta^k(fg)(0)$ of fg are nonnegative and log-concave without internal zeros.*

The proof, which is self-contained and works in the Newton basis, is given in Appendix B.

In the language of random sets: if S and T are independent random subsets of $[n]$ whose laws are invariant under permutations and whose cardinalities are ultra-log-concave of order n in the sense of Liggett [3], then so is $|S \cup T|$, since $\Pr(|S \cup T| = k) = \binom{n}{k}\kappa_k$ for $\Pr(S = A) = \rho_{|A|}$ and $\Pr(T = A) = \nu_{|A|}$. This closure theorem is Walkup's Theorem 4 for two sets [12]: his random sets are conditionally uniform, and their fundamental sequences are ρ and ν. He derives it from his Theorem 2, which proves that the double-binomial convolution $h_k = \sum_i \sum_r \binom{k}{i}\binom{i}{r}\rho_i \nu_{k-r}$, which is κ_k after the substitution $j = k - r$, preserves log-concavity; Proposition 5.3 expresses it in the Newton basis, and the proof in Appendix B is self-contained. Hoggar's theorem [2] is the analogue for the power basis. In terms of the linear map E with $E\binom{x}{j} = x^j$, the Newton coefficients of f are the coefficients of $E(f)$, and $E(fg)$ is Wagner's diamond product of $E(f)$ and $E(g)$ [1, (4.3)], [11, Theorem 2.3]; so Proposition 5.3 states that the diamond product preserves nonnegative coefficient sequences that are log-concave without internal zeros. For the numerator polynomial $W(f)$ of the generating function $\sum_{x\ge 0} f(x)t^x$, whose coefficients expand f in another basis, the pointwise product preserves real zeros of W by Wagner [11], ultra-log-concavity of its coefficients by Brändén, Ferroni and Jochemko [1], and their log-concavity without internal zeros by Liu and Mao [4].

Log-concave ratios on log-concave profiles. Corollary 5.2 extends to every log-concave Newton profile. Besides Proposition 5.3, the proof uses a sign criterion for quadratic forms in log-concave sequences.

Lemma 5.4. *Let $Q(u) = \sum_{0\le i\le j} q_{ij} u_i u_j$ with $q_{0j} \le 0$ for all j and $q_{ij} \ge 0$ for $2 \le i \le j$, and put $N_W(Q) = \sum_{i\le j,\, i+j=W} q_{ij}$. If $N_W(Q) \ge 0$ for every W, then $Q(u) \ge 0$ for every nonnegative sequence $(u_n)_{n\ge 0}$ that is log-concave without internal zeros.*

Proof. Fix W and put $p_i = u_i u_{W-i}$ for $0 \le i \le W/2$ and $p_{-1} = 0$. Log-concavity without internal zeros makes p nondecreasing. With $T_k = \sum_{k\le i\le W/2} q_{i,W-i}$ we get $\sum_{i\le W/2} q_{i,W-i} p_i = \sum_{k\le W/2} T_k(p_k - p_{k-1})$. Here $T_0 = N_W(Q) \ge 0$, $T_1 = N_W(Q) - q_{0W} \ge 0$, and $T_k \ge 0$ for $k \ge 2$. ∎

The weight sums are the coefficients of Q on geometric sequences: $Q(1, r, r^2, \ldots) = \sum_W N_W(Q) r^W$. This Abel argument, with the reduction to the full sum when the negative terms come first, is that of Wang and Yeh [13, Lemma 2.1 and Corollary 2.6].

Theorem 5.5 *(Log-concave ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros. Let $\rho(x) = \sum_k \rho_k \binom{x}{k}$, where $(\rho_k)_{k\ge 0}$ is nonnegative and log-concave without internal zeros and $\rho_0 > 0$. Put $a_i = 1 - d_i$, $b_i = (1-\rho(i))d_i$ and $c_i = \rho(i)d_i$. If b is nondecreasing, then the system has nonnegative linearization.*

This contains Theorem 4.1 and Corollary 5.2. It also contains the case $t^2 \ge r\tau$ of Theorem 4.2. The exponential ratios $\rho(x) = r(1+\varepsilon)^x$ are admitted on every log-concave Newton profile. The proof is computer-assisted in two places, both by exact rational arithmetic (listed in the supplement [9]):

- the signs of the coefficients in the edge bounds of step (vi), certified symbolically for all window sizes a;
- the key inequality (5.2) for windows $a \le 5$, where step (vii) uses exact linear-programming certificates.

Proof. Setup. Only $\rho_0, \dots, \rho_N$ enter, and truncating (ρ_k) after index N keeps it log-concave without internal zeros, so we may assume it finitely supported. Write $f_n(\ell) = \Delta^n d(\ell)$, $R_i = d_i/f_1(i)$, which is nondecreasing as in the proof of Theorem 4.1, and $P_k(x) = \Delta^k \rho(x) = \sum_h \binom{x}{h}\rho_{k+h}$. As in the proof of Theorem 4.2, $1 - \rho(m+1) \ge \Delta\rho(m)R_m$ for $0 \le m < N$, and $0 < \rho_0 \le \rho(i) \le 1$ for $i \le N$; ρ is nondecreasing.

For every integer $x \ge 0$, the sequence $(P_k(x))_k$ arises from (ρ_k) by the shifts of Lemma 2.1, so it is log-concave without internal zeros. Since $\rho(x+1)^2 - \rho(x)\rho(x+2) = P_1(x)^2 - P_0(x)P_2(x) \ge 0$, the positive sequence $(\rho(n))_{n\ge 0}$ is log-concave. (The function $x \mapsto \rho(x)$ of a real variable need not be: $\rho(x) = 1 + x/2 + x^2/2$ has $(\log\rho)''(0) > 0$.) Log-concavity in k gives $\rho(y)P_{k+1}(y) \le P_1(y)P_k(y)$. Combining these facts, for integers $0 \le x \le y$ and $k \ge 1$,

$$\rho(x)P_{k-1}(y+1) \le \rho(x+1)P_{k-1}(y), \qquad \rho(x)P_k(y) \le \Delta\rho(x)P_{k-1}(y). \tag{5.1}$$

Indeed, log-concavity of $x \mapsto \rho(x)$ gives $\rho(x)\rho(y+1) \le \rho(x+1)\rho(y)$, equivalently $\rho(x)P_1(y) \le \Delta\rho(x)\rho(y)$. For the first inequality, $\rho(y)P_{k-1}(y+1) = \rho(y)P_{k-1}(y) + \rho(y)P_k(y) \le \rho(y+1)P_{k-1}(y)$; multiply by $\rho(x)$ and divide by $\rho(y) > 0$. For the second, $\rho(x)\rho(y)P_k(y) \le \rho(x)P_1(y)P_{k-1}(y) \le \Delta\rho(x)\rho(y)P_{k-1}(y)$. By Proposition 5.3, the Newton coefficients of $c = \rho d$ are log-concave without internal zeros, and so is $(\varphi_n(\ell))_n$ for every ℓ. For $\ell + n \le N$ the Leibniz rule gives, all at ℓ,

$$\varphi_n = \sum_{j=0}^{n} \binom{n}{j} P_j(\ell+n-j) f_{n-j}, \qquad \psi_n = h_n - v_n - \widehat{w}_n,$$

with $h_n = (1-\rho(\ell+n))f_n$, $v_n = n\Delta\rho(\ell+n-1)f_{n-1}$ and $\widehat{w}_n = \sum_{j=2}^{n} w_n^{(j)}$, where $w_n^{(j)} = \binom{n}{j}P_j(\ell+n-j)f_{n-j}$. Here φ_n, h_n, v_n, $w_n^{(j)}$ and $\widehat{w}_n$ are nonnegative, while ψ_n need not be.

Moves and detours. Split the moves as in the proof of Theorem 4.2, with W_j now formed from $\widehat{w}_n$. Lemma 2.1 for φ gives $B_0(\ell,a)B_{j-1}(\ell+1,a-2) \ge 4c_{\ell+1}B_j(\ell,a)$ for $j \ge 1$, as

in [5, Lemma 3.4]. The detours of H and C', with constants 10 and 6, hold verbatim: they used only $\varphi_2(\ell) \geq \rho(\ell+1)f_2(\ell)$, Lemma 2.1 for f, and $0 \leq \rho \leq 1$ nondecreasing.

For V, the term $\rho(\ell+2)f_2$ of φ_2, the first inequality of (5.1) with $k=2$, $x=\ell+1$, $y=\ell+n$, and Lemma 2.1 for f with indices $(2,n-1)$ give

$$\varphi_2(\ell)v_n(\ell+1) \geq \tfrac{n}{n+2}\, c_{\ell+1}v_{n+2}(\ell).$$

For $\widehat{w}$, expand $\varphi_2(\ell) = \rho(\ell+2)f_2 + 2\Delta\rho(\ell+1)f_1 + P_2(\ell)f_0$, and pair its three terms with the components of $\widehat{w}_n(\ell+1)$:

- $\rho(\ell+2)f_2$ with component $i \leq n$, by the first inequality of (5.1) and indices $(2,n-i)$, gives at least $\binom{n}{i}\binom{n+2}{i}^{-1} c_{\ell+1}w^{(i)}_{n+2}(\ell)$;
- $2\Delta\rho(\ell+1)f_1$ with component $i \leq n$, by the second inequality of (5.1) and indices $(1,n-i)$, gives at least $2\binom{n}{i}\binom{n+2}{i+1}^{-1} c_{\ell+1}w^{(i+1)}_{n+2}(\ell)$;
- $P_2(\ell)f_0$ with component n gives at least $c_{\ell+1}w^{(n+2)}_{n+2}(\ell)$, since $P_2(\ell)P_n(\ell+1) \geq (P_0+P_1)(\ell)P_{n+2}(\ell)$ by log-concavity in k, and $f_0(\ell+1) = d_{\ell+1}$.

When $n-i=0$ the index condition is trivial. Collecting by target,

$$\varphi_2(\ell)\widehat{w}_n(\ell+1) \geq c_{\ell+1}\sum_{i=2}^{n+2}\theta_{n,i}w^{(i)}_{n+2}(\ell),$$

where $\theta_{n,2} = \binom{n}{2}\binom{n+2}{2}^{-1}$, $\theta_{n,i} = 1 - \frac{i(i-1)}{(n+1)(n+2)}$ for $3 \leq i \leq n$, $\theta_{n,n+1} = \frac{2}{n+2}$ and $\theta_{n,n+2} = 1$.

As in the proof of Theorem 4.2, the coefficient quotients are $\frac{j(m-1)(m-2)}{2(m-j-1)}$ for V, and $\theta_{m-2,i}\frac{jm(m-1)}{2(m-j-1)}$ for $\widehat{W}$, on $m \geq 2j+1$, $j \geq 2$. The first is at least 6, by $(m-4)(m-5) \geq 0$ at $j=2$. The second exceeds 2: it is at least $\frac{j(m-2)(m-3)}{2(m-j-1)} \geq m-2$ for $i=2$, at least $\frac{j(2m-3)}{m-j-1}$ for $3 \leq i \leq m-2$, equal to $\frac{j(m-1)}{m-j-1}$ for $i=m-1$, and at least 10 for $i=m$. Hence

$$B_0V'_{j-1} \geq 6c_{\ell+1}V_j \qquad \text{and} \qquad B_0W'_{j-1} \geq 2c_{\ell+1}W_j \qquad (j \geq 2).$$

Resummation. We prove below, for $(\ell,a) \in \Omega$ with $a \geq 3$,

$$A^{\psi}_0(\ell,a)B_0(\ell+1,a-1) + B_0(\ell,a)A^{\psi}_0(\ell+1,a-2) \geq c_{\ell+1}\big(4H_1 - 2V_1\big)(\ell,a). \tag{5.2}$$

With the notation of the proof of Theorem 4.2, the inequalities above give $P^2 \geq 4Q$, $H_+P \geq 10H_-$, $V_+P \geq 6V_-$, $W_+P \geq 2W_-$, $C_+P \geq 6C_-$ and $A_0P + PA_0 + 2V_1 \geq 4H_1$. Since $(I - P/\mu)G = (1-\frac{2}{\mu})G + \frac{2}{\mu}I$, one has $G(H_+ - H_-)G \geq 0$, $GCG \geq 0$, $G(W_+ - W_-)G \geq 0$ and $G(V_+ - V_-)G \geq \frac{2}{3}GV_1G$, so

$$GAG \geq G\big(A_0 - \tfrac{1}{4}(A_0P + PA_0) + \tfrac{1}{6}V_1\big)G = \tfrac{1}{2}(GA_0 + A_0G) + \tfrac{1}{6}GV_1G \geq 0,$$

and (P) follows as before.

Proof of (5.2): notation. Use the notation of the proof of Theorem 4.1: $u_n = f_n(\ell)$, which is log-concave without internal zeros, y_n, D, X, X', Y, Y', J, H_*. Put also $A = d_{\ell+a}$, $B = d_{\ell+a-1}$, $F = A - B = f_1(\ell+a-1)$, $X'' = B - d_\ell$ and $P_k = P_k(\ell)$. Since $c_{\ell+n} = d_{\ell+n}\sum_k \binom{n}{k}P_k$ for $n \leq a$, the following are linear in P:

- $B_0(\ell,a) = \sum_k \beta_k P_k$ and $B_0(\ell+1,a-1) = \sum_k \beta'_k P_k$, where
$$\beta_0 = Y, \quad \beta_1 = a(A-D), \quad \beta_k = \tbinom{a}{k}A \ (k \geq 2),$$
$$\beta'_0 = Y', \quad \beta'_1 = aA - 2(a-1)d_{\ell+2} + (a-2)D,$$
$$\beta'_2 = \tbinom{a}{2}A - (a-1)d_{\ell+2}, \quad \beta'_k = \tbinom{a}{k}A \ (k \geq 3);$$
- $c_{\ell+a} - c_\ell = \sum_k \chi_k P_k$ and $c_{\ell+a-1} - c_{\ell+1} = \sum_k \chi'_k P_k$;
- $c_{\ell+a} - c_{\ell+a-1} = \sum_k \delta_k P_k$ with $\delta_k = \binom{a}{k}F + \binom{a-1}{k-1}B$;
- $H_1 = H_* - \sum_k \zeta_k P_k$ with $\zeta_k = \sum_{n\geq 3}(n-2)\binom{n}{k}y_n$, and $V_1 = \sum_k \nu_k P_k$ with $\nu_k = \sum_{n\geq 3}(n-2)n\binom{n-1}{k-1}\binom{a}{n}u_{n-1}$;
- $c_{\ell+1} = D(P_0+P_1)$.

As $A_0^\psi(\ell,a) = X - (c_{\ell+a} - c_\ell)$ and $A_0^\psi(\ell+1,a-2) = X' - (c_{\ell+a-1} - c_{\ell+1})$, the difference of the two sides of (5.2) is $\mathrm{Key}(P) = \mathrm{Lin}(P) - \mathrm{Quad}(P)$. Here $\mathrm{Lin}(P) = \sum_k l_k P_k$ with $l_k = X\beta'_k + \beta_k X' - 4[k \leq 1]DH_*$, and Quad is a quadratic form. We use repeatedly:

$$AX'' \geq (a-1)d_\ell F, \qquad AX' \geq (a-2)DF, \qquad u_0 F \leq u_1 B, \qquad FD \leq u_1 A. \tag{5.3}$$

The first two hold because $A/F \geq R_{\ell+a-1} \geq R_{\ell+i}$ gives $(A/F)f_1(\ell+i) \geq d_{\ell+i}$ for $i \leq a-1$; summing over $0 \leq i \leq a-2$, respectively $1 \leq i \leq a-2$, gives them. The last two follow from $u_0 u_{m+1} \leq u_1 u_m$, as $F = \sum_m \binom{a-1}{m}u_{m+1}$, $B = \sum_m \binom{a-1}{m}u_m$ and $D = u_0 + u_1$.

(i) Reduction to a quadratic form in P. By monotonicity of b at $\ell+a-1$, $\Delta c := c_{\ell+a} - c_{\ell+a-1} \leq F$. With $\Phi(P) := \Delta c \cdot \mathrm{Lin}(P) - F\,\mathrm{Quad}(P)$ one has $\mathrm{Key} = (1 - \Delta c/F)\mathrm{Lin} + \Phi/F$, so it suffices that every $l_k \geq 0$ and $\Phi \geq 0$. With $\chi_k = \binom{a}{k}A - [k=0]d_\ell$ and $\chi'_k = \binom{a-1}{k}B - \binom{1}{k}D$, so that $E^1_m = X\delta_m - F\chi_m$, $E^2_m = X'\delta_m - F\chi'_m$ and $E^3_m = 2F\nu_m - 4(H_*\delta_m - F\zeta_m)$, expansion gives $\Phi(P) = \sum_{k,m} M_{k,m}P_kP_m$ with

$$M_{k,m} = \beta'_k E^1_m + \beta_k E^2_m + [k \leq 1]D\,E^3_m,$$

and explicitly $E^1_0 = E^2_0 = E^3_0 = 0$ and, for $m \geq 1$,

$$E^1_m = \tbinom{a-1}{m-1}AX'' - \tbinom{a-1}{m}d_\ell F, \qquad E^2_m = \tbinom{a-1}{m-1}AX' - \Big(\tbinom{a-1}{m} - [m=1]\Big)DF,$$
$$E^3_m = 2F\nu_m - 4\tbinom{a-1}{m-1}BH_* - 4FZ_m, \qquad Z_m = \sum_{n\geq 3}(n-2)\Big(\tbinom{a}{m} - \tbinom{n}{m}\Big)y_n.$$

Since (P_k) is log-concave without internal zeros, the Abel summation in the proof of Lemma 5.4, applied to every anti-diagonal of Φ, shows $\Phi \geq 0$ if every central sum $S_{\sigma,i} = \sum_{k=i}^{\sigma-i} M_{k,\sigma-k}$, $0 \leq i \leq \sigma/2$, is nonnegative. As $M_{k,0} = 0$, for $\sigma \geq 3$

$$S_{\sigma,0} = S_{\sigma,1} + M_{0,\sigma} \qquad \text{and} \qquad S_{\sigma,1} = S_{\sigma,2} + M_{1,\sigma-1} + M_{\sigma-1,1}.$$

Since $\binom{a-1}{m} = \frac{a-m}{m}\binom{a-1}{m-1}$ and $\frac{a-m}{m(a-1)} \leq \frac{a-m}{m(a-2)} \leq \frac{1}{2}$ for $m \geq 2$, (5.3) gives, for $m \geq 2$,

$$E^1_m \geq \tfrac{1}{2}\tbinom{a-1}{m-1}AX'' \geq 0 \qquad \text{and} \qquad E^2_m \geq \tfrac{1}{2}\tbinom{a-1}{m-1}AX' \geq 0.$$

Also every $\beta_k \ge 0$ and $\beta'_k \ge 0$; for β'_2 use $\binom{a}{2}A \ge (a-1)d_{\ell+2}$. Finally $E^1_1 = AX'' - (a-1)d_\ell F \ge 0$ and $E^2_1 = AX' - (a-2)DF \ge 0$.

(ii) The central sums with $i \ge 2$, and the coefficients l_k. For $k, m \ge 2$, $M_{k,m} = \beta'_k E^1_m + \beta_k E^2_m \ge 0$, so $S_{\sigma,i} \ge 0$ for $i \ge 2$. Next, $l_0 = K \ge 0$ by the proof of Theorem 4.1, and $l_k \ge 0$ for $k \ge 2$ termwise. By convexity, $A - D \ge (a-1)f_1(\ell+1) = (a-1)(d_{\ell+2} - D)$, so $\beta'_1 \ge (a-2)(A-D)$ and

$$l_1 \ge L := (A-D)\big((a-2)X + aX'\big) - 4DH_*.$$

(iii) The central sums with $\sigma \le 2$. Take $P = (P_0, P_1, 0, \dots)$ with $P_0 \ge 0$ and $P_1 > 0$, and put $\lambda = F/\Delta c(P)$. Then $\mathrm{Key}(\lambda P) = (\lambda/\Delta c)\Phi(P)$, and λP is the linear ratio $\tilde\rho(\ell + x) = \lambda(P_0 + P_1 x)$. Its $\tilde b = (1-\tilde\rho)d$ satisfies $\Delta\tilde b(\ell + a - 1) = 0$. So $\alpha := 1 - \tilde\rho(\ell + a) = tR_{\ell+a-1} \ge tu_0/u_1$ with $t = \lambda P_1$. For such α and t, the proof of Theorem 4.1 shows (4.2). Its two weights A^ψ_0 are nonnegative, since $X \ge au_1$ and $X' \ge (a-2)u_1$, and $B_0(\ell, a) \ge \tilde\rho(\ell+1)Y$, $B_0(\ell+1, a-1) \ge \tilde\rho(\ell+1)Y'$, since $\beta_1 \ge Y$ and $\beta'_1 \ge Y'$. Hence $\mathrm{Key}(\lambda P) \ge 0$, that is, $\Phi(P) = P_1(M_{0,1}P_0 + M_{1,1}P_1) \ge 0$. This gives $S_{1,0} = M_{0,1} \ge 0$ and $S_{2,1} = M_{1,1} \ge 0$. The sum $S_{2,0} = M_{0,2} + M_{1,1}$ is treated in (v).

(iii′) $M_{1,m} + M_{m,1} \ge 0$ for $m \ge 2$ and $a \ge 6$. Put $c'_m = \binom{a-1}{m-1}$. Then

$$M_{1,m} + M_{m,1} = \big[\beta_1 E^2_m - 4c'_m DBH_*\big] + \big[\beta'_1 E^1_m - DF(4Z_m - 2\nu_m)\big] + \beta'_m E^1_1 + \beta_m E^2_1,$$

and the last two terms are nonnegative.

- First bracket: $E^2_m \ge \frac12 c'_m AX'$ and $A \ge B$, so it is nonnegative if $R := a(A-D)X' - 8DH_* \ge 0$.
- Second bracket: $Z_m \le \binom{a}{m}H_* \le \frac{a}{2}c'_m H_*$, $\nu_m \ge 0$, $F \le A$, $E^1_m \ge \frac12 c'_m AX''$ and $\beta'_1 \ge (a-2)(A-D)$. So it is nonnegative if $R_2 := (a-2)(A-D)X'' - 4aDH_* \ge 0$.

(iv) $M_{0,s} \ge 0$ for $s \ge 3$. Put $c' = \binom{a-1}{s-1}$ and $c'' = \binom{a-1}{s}$. Then

$$M_{0,s} = Y'E^1_s + YE^2_s + DE^3_s = c'BK'' + FG_s,$$

where

$$K'' = Y'X'' + YX' - 4DH_*, \quad G_s = c'(Y'X'' + YX') - c''(d_\ell Y' + DY) + D(2\nu_s - 4Z_s).$$

Note $u_1 = d_{\ell+1} - d_\ell > 0$. If $K'' \ge 0$, then $u_1 B \ge u_0 F$ gives $u_1 M_{0,s} \ge F(c'u_0K'' + u_1G_s)$, and

$$c'u_0K'' + u_1G_s = D\,Q_s + c''u_1^2Y',$$
$$Q_s = c'(Y'X'' + YX') - 4c'u_0H_* + u_1(2\nu_s - 4Z_s) - c''u_1(Y + Y').$$

So $M_{0,s} \ge 0$ once $K'' \ge 0$ and $Q_s \ge 0$.

(v) $S_{2,0} \ge 0$ for $a \ge 5$. With $c'' = \binom{a-1}{2}$, the definitions give the identity

$$S_{2,0} = \beta'_1 E^1_1 + \big[\beta_1 E^2_1 - c''F(d_\ell Y' + DY)\big] + B\,K_a + F\,W,$$

where $K_a = (a-1)K'' - 4DH_*$ and $W = (a-1)(Y'X'' + YX') + D(2\nu_1 + 2\nu_2 - 4Z_1 - 4Z_2)$.

- Since $\beta_1 = a(X'+F)$, $E_1^2 = AX' - (a-2)DF$, $d_\ell \le D$ and $FD \le u_1 A$, the bracket is at least $A\Psi$, where $\Psi = a(A-D)J - c''u_1(Y+Y')$. Recall $J = X' - (a-2)u_1$ and $X' + F = A - D$.
- If $K_a \ge 0$, then $u_1(BK_a + FW) \ge F(u_0 K_a + u_1 W) = FD\widehat{Q}$, where $\widehat{Q} = (a-1)(Y'X'' + YX') - 4au_0H_* + u_1(2\nu_1 + 2\nu_2 - 4Z_1 - 4Z_2)$.

Hence $u_1 S_{2,0} \ge u_1\beta_1' E_1^1 + u_1 A\Psi + FD\widehat{Q} \ge 0$ once K_a, Ψ and $\widehat{Q}$ are nonnegative.

(vi) Edge bounds. Steps (ii)–(v) reduce (5.2) to the nonnegativity of the quadratic forms K, L, R, R_2, K'', Q_s $(3 \le s \le a)$, K_a, Ψ and $\widehat{Q}$ in the sequence u. For Q_s, first remove the dependence on s: by $\nu_s \ge 0$, $Z_s \le \binom{a}{s} H_*$, $\binom{a}{s} \le \frac{a}{3}\binom{a-1}{s-1}$ and $\binom{a-1}{s} \le \frac{a-3}{3}\binom{a-1}{s-1}$ for $s \ge 3$, one has $Q_s \ge \binom{a-1}{s-1} Q^*$ with

$$Q^* = Y'X'' + YX' - 4u_0H_* - \tfrac{4a}{3}u_1H_* - \tfrac{a-3}{3}u_1(Y+Y').$$

Each of these forms is a sum of products xy of linear forms $x = \sum_{i\ge1} x_i u_i$ and $y = \sum_{j\ge1} y_j u_j$ with nonnegative coefficients, plus u_1 times a linear form, minus u_0 times a linear form with nonnegative coefficients. Since u is log-concave without internal zeros, $u_i u_j \ge u_1 u_{i+j-1}$ for $i, j \ge 1$ and $u_0 u_{n+1} \le u_1 u_n$. Keeping in xy only the pairs with $i \le 2$ or $j \le 2$ gives

$$xy \ge u_1 \sum_n e_n(x,y) u_n, \qquad e_n(x,y) = x_1 y_n + x_2 y_{n-1} + [n \ge 3]\, y_1 x_n + [n \ge 4]\, y_2 x_{n-1},$$

and $u_0 \sum_{n\ge1} h_n u_n \le u_1 \sum_{n\ge0} h_{n+1} u_n$. So each form is at least $u_1 \sum_n e_n u_n$ with explicit coefficients e_n, and it suffices that every $e_n \ge 0$.

- For $4 \le n \le a$ write $n = 4 + p$ and $a = n + k$. Then $e_n / \binom{a}{n}$ is a rational function of (p, k) whose numerator has nonnegative coefficients and whose denominator is strictly positive for $p, k \ge 0$.
- For $n \le 3$, e_n is a polynomial in a whose coefficients are nonnegative after $a \mapsto a + a_0$, with $a_0 = 6$ for R_2, $a_0 = 5$ for K_a and $a_0 = 3$ otherwise.
- For $n \ge a + 1$ only nonnegative products remain.

These certificates are formalized in Lean, and `runs/lean_plan/edge_exact.py` checks every e_n in exact arithmetic for $a \le 60$.

(vii) Small windows. Steps (iii′) and (v) need $a \ge 6$ and $a \ge 5$. For $3 \le a \le 5$, each sum $M_{1,m} + M_{m,1}$ $(2 \le m \le a)$ and, for $a \le 4$, $S_{2,0}$ is a cubic form in u that is an exact nonnegative rational combination of the products $u_k(u_{i+1}u_{j-1} - u_i u_j)$ $(i + 1 \le j - 1)$ and of monomials (`runs/lean_plan/window_cert.py` in [7]). This proves (5.2) and the theorem. ∎

Certificate map. Steps (i)–(v) of this proof, and the corresponding reductions in the proof of Theorem 5.6 below, hold for every N and use no computer assistance. The remaining obligations, and the Lean declarations that prove them for every N (prefix `Szkraw.` omitted), are these.

- Theorem 5.5, step (vi): the edge bounds, *symbolic* for every window size $a \geq 3$ (`eK_nonneg`, `eL1_nonneg`, `eR_nonneg`, `eR2_nonneg`, `eKpp_nonneg`, `eQs_nonneg`, `eKa_nonneg`, `ePsi_nonneg`, `eQhat_nonneg`).
- Theorem 5.5, step (vii): the windows $3 \leq a \leq 5$, *finite*, one exact certificate for each case (`win_MM1_3_2`, ..., `win_MM1_5_5`, `win_S20_3`, `win_S20_4`).
- Theorem 5.6: the tail sums of (5.5), (5.6) and (5.4), *symbolic* in ℓ and n (`cubicB_gen`, `cubicB_two`, ..., `cubicB_five`, `cubicB_corner`, `cubicE_gen`, `cubicE_three`, ..., `cubicE_six`), and $\Phi_3^{\mathrm{loc}} \geq 0$, *symbolic* in a (`Phi3_nonneg`).
- Corollary 5.7: the modified coefficient in (5.5) for $n \leq 5$, *symbolic* in ℓ; Lean proves it within the declarations for (5.5) above, which are stated under $r\sigma \leq \frac{7}{4}t\tau$.

A symbolic certificate has coefficients that are polynomials or rational functions of the parameters and covers the whole range. The supplement [9] lists the certificate scripts and the exact cross-checks, which recompute coefficients at finitely many sizes.

A worked certificate. The smallest obligation shows how a step of the proof becomes a polynomial identity checked by Lean. In step (vii) with $a = 3$, the central sum $S_{2,0} = M_{0,2} + M_{1,1}$ must be nonnegative. Substituting the definitions of step (i), with $u_n = f_n(\ell)$, gives a cubic form in $u_0, \dots, u_3$ with integer coefficients,

$$\begin{aligned} S_{2,0} \;=\; & 24u_1^3 + 135u_1^2u_2 + 189u_1u_2^2 + 66u_2^3 + 21u_1^2u_3 + 98u_1u_2u_3 + 52u_2^2u_3 \\ & + 11u_1u_3^2 + 10u_2u_3^2 + 12u_0u_1u_2 + 12u_0u_2^2 - 42u_0u_1u_3 - 29u_0u_2u_3 \\ & - 11u_0u_3^2 - 12u_0^2u_3, \end{aligned}$$

and the obligation is $S_{2,0} \geq 0$ for every nonnegative sequence u that is log-concave without internal zeros. For such u the products $u_k(u_{i+1}u_{j-1} - u_iu_j)$ with $i+1 \leq j-1$ are nonnegative, and so are all monomials. The script `runs/lean_plan/window_cert.py` solves an exact linear program for nonnegative rational multipliers of these generators and returns

$$\begin{aligned} S_{2,0} \;=\; & 36u_0(u_1u_2 - u_0u_3) + 42u_0(u_2^2 - u_1u_3) + 24u_1(u_1^2 - u_0u_2) + 189u_1(u_2^2 - u_1u_3) \\ & + 135u_2(u_1^2 - u_0u_2) + 29u_3(u_1^2 - u_0u_2) + 98u_3(u_1u_2 - u_0u_3) \\ & + 24u_0^2u_3 + 105u_0u_2^2 + 87u_0u_3^2 + 181u_1^2u_3 + 11u_1u_3^2 + 66u_2^3 + 52u_2^2u_3 + 10u_2u_3^2, \end{aligned}$$

an identity of polynomials, which the script checks by expansion. The Lean statement `Szkraw.win_S20_3` asserts $S_{2,0} \geq 0$ for every such sequence. Its proof records the fifteen generators above as nonnegative facts, expands $S_{2,0}$ from the Lean definitions of the $M_{k,m}$, and closes the goal by linear arithmetic, which the Lean kernel checks. The symbolic certificates of step (vi) use the transfers $u_iu_j \geq u_1u_{i+j-1}$ and $u_0u_{n+1} \leq u_1u_n$ in place of these generators, and their coefficients e_n are rational functions of a and n whose numerators have nonnegative coefficients, so one certificate covers every window size.

For cubic ratios, log-concavity of the Newton coefficients can be replaced by a single inequality.

Theorem 5.6 *(Cubic ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, and let $\rho(x) = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$ with $r > 0$, $t, \tau, \sigma \geq 0$ and $r\sigma \leq t\tau$. Put $a_i = 1 - d_i$, $b_i = (1 - \rho(i))d_i$ and $c_i = \rho(i)d_i$. If b is nondecreasing, then the system has nonnegative linearization.*

If (r, t, τ, σ) is log-concave without internal zeros, then $r\sigma \leq t\tau$: for $t\tau > 0$ this follows from $t^2\tau^2 \geq r\tau \cdot t\sigma$, and otherwise $\sigma = 0$. Theorem 5.6 also admits cubic ratios with $t^2 < r\tau$ or $\tau^2 < t\sigma$, which Theorem 5.5 does not cover. The failing cubic examples of Appendix A violate $r\sigma \leq t\tau$. The proof is computer-assisted: the inequalities (5.5), (5.6) and (5.4) and the inequality $\Phi_3^{\mathrm{loc}} \geq 0$ below are reduced to families of explicit polynomials with nonnegative coefficients.

Proof. Setup. As in the proof of Theorem 4.2, $R_i = d_i/f_1(i)$ is nondecreasing, $1 - \rho(m+1) \geq \Delta\rho(m)R_m$ for $0 \leq m < N$, and ρ is nondecreasing with $0 < \rho(i) \leq 1$ for $i \leq N$; here $\Delta\rho(x) = t + \tau x + \sigma\binom{x}{2}$ and $\Delta^2\rho(x) = \tau + \sigma x$. For $\ell + n \leq N$ the Leibniz rule gives, all at ℓ,

$$\varphi_n = \rho(\ell+n)f_n + n\Delta\rho(\ell+n-1)f_{n-1} + \tbinom{n}{2}\Delta^2\rho(\ell+n-2)f_{n-2} + \tbinom{n}{3}\sigma f_{n-3}.$$

Since $\binom{\ell+n-1}{2} = \binom{\ell}{2} + \ell(n-1) + \binom{n-1}{2}$ and $n\binom{n-1}{2} = \binom{n}{2}(n-2) = 3\binom{n}{3}$, this gives $\psi_n = h_n - v_n - e_n$ with $h_n = (1 - \rho(\ell+n))f_n$,

$$v_n = n\Gamma_n(\ell)f_{n-1}, \qquad \Gamma_n(\ell) = t + \tau(\ell+n-1) + \sigma\Big(\tbinom{\ell}{2} + \ell(n-1)\Big),$$
$$e_n = \tbinom{n}{2}(\tau + \sigma\ell)f_{n-2} + \sigma\tbinom{n}{3}\big(3f_{n-1} + 3f_{n-2} + f_{n-3}\big),$$

all nonnegative. Write $u_k = f_k(\ell)$, so that $f_k(\ell+1) = u_k + u_{k+1}$; then (u_k) is log-concave, and $u_0 \leq \ell u_1$ by convexity.

Moves. Split the moves as in the proof of Theorem 4.2, with V_j formed from v_n and with a third family $E_j = W_{2j+1,j}$ formed from e_n in place of W_j.

Detours of H, V and C'. The detours of H and C' hold verbatim, with constants 10 and 6. For V,

$$(n+1)\Gamma_n(\ell+1) - n\Gamma_{n+2}(\ell) = t + \tau\ell + \sigma\Big(\tbinom{\ell}{2} + n^2 - 1\Big) \geq 0,$$

so Lemma 2.1 with indices $(2, n-1)$ gives $(n+1)(n+2)\varphi_2 v'_n \geq n^2 c_{\ell+1}v_{n+2}$ for $n \geq 3$, and $B_0V'_{j-1} \geq \frac{9}{2}c_{\ell+1}V_j$ for $j \geq 2$ as before. (With the full σ-part $n\sigma\binom{\ell+n-1}{2}f_{n-1}$ in v_n, the corresponding difference is negative for $\ell < n+1$; without any σ-part in v_n, the reduction of the key inequality below fails for large ℓ.)

Detour of E. We show, for $n \geq 3$ and $\ell + n + 2 \leq N$,

$$(n+1)(n+2)\Big(\varphi_2(\ell) + \frac{n}{3}\varphi_3(\ell)\Big)e_n(\ell+1) \geq 3(n-1)\,c_{\ell+1}e_{n+2}(\ell). \tag{5.4}$$

Granting this, $B_0(\ell, a) \geq \binom{a}{2}\varphi_2(\ell) + \binom{a}{3}\varphi_3(\ell)$, $\binom{a}{3}\binom{a-2}{n} = \frac{a-2}{3}\binom{a}{2}\binom{a-2}{n}$ with $a - 2 \geq n$ whenever $\binom{a}{n+2} \neq 0$, and $\binom{a}{2}\binom{a-2}{n} = \binom{a}{n+2}\binom{n+2}{2}$ give, for $j \geq 2$,

$$B_0(\ell, a)E_{j-1}(\ell+1, a-2) \geq \tfrac{3}{2}c_{\ell+1}\sum_{m \geq 2j+1}(m-3)\tbinom{m-j-2}{j-1}\tbinom{a}{m}e_m(\ell) \geq 3c_{\ell+1}E_j(\ell, a),$$

because $(m-3)\binom{m-j-2}{j-1} - 2\binom{m-j-1}{j} = \binom{m-j-2}{j-1}\frac{(j-2)(m-1)}{j} \geq 0$.

Detour of B. We show, for $n \geq 2$, $\ell + n + 2 \leq N$ and $(\ell, n) \neq (0, 2)$,

$$(n+1)(n+2)\,\varphi_2(\ell)\varphi_n(\ell+1) \;\geq\; 8(n-1)\,c_{\ell+1}\varphi_{n+2}(\ell), \tag{5.5}$$

and, if $N \geq 4$,

$$\big(12\varphi_2(0) + 8\varphi_3(0)\big)\varphi_2(1) \;\geq\; 8c_1\varphi_4(0). \tag{5.6}$$

Granting these, the argument after (4.3) gives $B_0(\ell,a)B_{j-1}(\ell+1,a-2) \geq 4c_{\ell+1}B_j(\ell,a)$ for $j \geq 1$. The only exception is the term $m = 4$ for $\ell = 0$, $j = 1$. There $B_0(0,a) \geq \binom{a}{2}\varphi_2(0) + \binom{a}{3}\varphi_3(0)$, $\binom{a}{2}\binom{a-2}{2} = 6\binom{a}{4}$ and $\binom{a}{3}\binom{a-2}{2} = 2(a-2)\binom{a}{4} \geq 4\binom{a}{4}$ reduce the term to (5.6). Inequality (5.5) fails at $(\ell, n) = (0, 2)$ in general.

Proof of (5.5), (5.6) and (5.4). In each, the difference of the two sides is a quadratic form $\sum \kappa_{\beta\gamma}\rho_\beta\rho_\gamma$ in $(\rho_0, \rho_1, \rho_2, \rho_3) = (r, t, \tau, \sigma)$. Its coefficients are bilinear forms in the u_k, with coefficients polynomial in ℓ and n.

- Since $t\tau\kappa_{t\tau} + r\sigma\kappa_{r\sigma} = (t\tau - r\sigma)\kappa_{t\tau} + r\sigma(\kappa_{t\tau} + \kappa_{r\sigma})$ and $r\sigma \leq t\tau$, the form is nonnegative if $\kappa_{t\tau}$, $\kappa_{t\tau} + \kappa_{r\sigma}$ and all coefficients other than $\kappa_{r\sigma}$ are nonnegative as forms in u.

- For (5.5) the coefficients κ_{rr}, $\kappa_{r\tau}$, $\kappa_{\tau\tau}$ need not be checked: the form with $t = \sigma = 0$ is (4.3) for the ratio $r + \tau\binom{x}{2}$. The same holds for (5.6), whose left side exceeds that of (5.5) at $(0, 2)$ by $8\varphi_3(0)\varphi_2(1) \geq 0$.

- Every remaining coefficient is a sum $\sum c_{ij}u_iu_j$ over $i \leq j$. Group its terms by $i + j = m$. Within a group, u_iu_{m-i} is nondecreasing in i by log-concavity, so with the tail sums $T_k = \sum_{i \geq k} c_{i,m-i}$ and the smallest index k_0 of the group,

  $$\sum_i c_{i,m-i}u_iu_{m-i} \;=\; T_{k_0}u_{k_0}u_{m-k_0} + \sum_{k>k_0} T_k\,(u_ku_{m-k} - u_{k-1}u_{m-k+1}),$$

  and the coefficient is nonnegative if all $T_k \geq 0$.

- The tail sums are polynomials in (ℓ, n). For (5.5) with $n \geq 6$ and for (5.4) with $n \geq 7$ they have nonnegative coefficients in $(\ell, n-6)$, respectively $(\ell, n-7)$. For each n in $3 \leq n \leq 5$ ((5.5)) and $3 \leq n \leq 6$ ((5.4)) they are polynomials in ℓ with nonnegative coefficients. The same holds for (5.5) with $n = 2$ and $\ell \geq 1$ after $\ell \mapsto \ell + 1$, and for (5.6), where $u_0 = 0$.

The scripts `runs/lean_plan/gen_cubicB.py` and `runs/lean_plan/gen_cubicE.py` check every tail-coefficient sign in exact rational arithmetic and generate the Lean certificates; Lean checks the certificate identities.

Key inequality. We show, for $(\ell, a) \in \Omega$ with $a \geq 3$,

$$A_0^\psi(\ell,a)B_0(\ell+1,a-1) + B_0(\ell,a)A_0^\psi(\ell+1,a-2) \;\geq\; c_{\ell+1}\big(4H_1 - 2V_1 - E_1\big)(\ell,a). \tag{5.7}$$

Use the notation of the proof of Theorem 4.2, and put $P_1 = \Delta\rho(\ell)$ and $P_2 = \Delta^2\rho(\ell) = \tau + \sigma\ell$, so that $\rho(\ell+k) = \rho(\ell) + kP_1 + \binom{k}{2}P_2 + \binom{k}{3}\sigma$. Put $\alpha = 1 - \rho(\ell+a)$, $s_q = P_1 +$

$(a-1)P_2$ and $s = \Delta\rho(\ell+a-1) = s_q + \binom{a-1}{2}\sigma$. Then $\alpha \ge su_0/u_1$ and $s_q \ge (a-1)P_2$. Backward differences give

$$\begin{aligned} A_0^{\psi}(\ell,a) &= \alpha X - s_q a u_0 + P_2\tbinom{a}{2}u_0 - \sigma\tbinom{a}{3}u_0,\\ A_0^{\psi}(\ell+1,a-2) &= (\alpha+s)X' - s_q(a-2)D + P_2\tbinom{a-1}{2}D - \sigma\tbinom{a-1}{3}D, \end{aligned}$$

and, reindexing as before,

$$4H_1 - 2V_1 - E_1 = 4\alpha H_* + 2s_q T + 4P_2 Z - P_2\sum_{n\ge1} n\tbinom{a-n}{2}y_n + \sigma(4T_3 - E_3),$$

where $T_3 = \sum_{n\ge3}(n-2)\big(\binom{a}{3} - \binom{n}{3}\big)y_n$ and $E_3 = \sum_{n\ge3}(n-2)\binom{n}{3}\binom{a}{n}(3u_{n-1} + 3u_{n-2} + u_{n-3})$. With $B_0(\ell,a) \ge pY$, $B_0(\ell+1,a-1) \ge pY'$ and $c_{\ell+1} = pD$, it suffices that $R = A_0^{\psi}(\ell,a)Y' + YA_0^{\psi}(\ell+1,a-2) - D(4H_1 - 2V_1 - E_1) \ge 0$. R is affine in α, with coefficient $XY' + YX' - 4DH_*$. Its part without σ is the expression bounded in the proof of Theorem 4.2, with s_q and P_2 in place of s and τ; that bound holds for every value α' of α with $\alpha' u_1 \ge s_q u_0$. Apply it with $\alpha' = \alpha - \binom{a-1}{2}\sigma z$, where $z = u_0/u_1$. This gives $R \ge \sigma\Phi_3^{\mathrm{loc}}$ with

$$\Phi_3^{\mathrm{loc}} = \tbinom{a-1}{2}z\,(XY' + YX' - 4DH_*) - \tbinom{a}{3}u_0Y' + Y\Big(\tbinom{a-1}{2}X' - \tbinom{a-1}{3}D\Big) - D(4T_3 - E_3).$$

$\Phi_3^{\mathrm{loc}} \ge 0$. Write $E_3 = \binom{a}{3}u_0 + E_3^0$ and $H_*^+ = \sum_{n\ge2}(n-1)\binom{a}{n+1}u_n$, so that $u_0H_* \le u_1H_*^+$. Bounding the only negative term quadratic in u_0, $-4\binom{a-1}{2}u_0^2H_*$, this way and dropping $\binom{a}{3}u_0^2u_1 \ge 0$ gives $u_1\Phi_3^{\mathrm{loc}} \ge u_1A + u_0B$ with

$$\begin{aligned} A &= \tbinom{a-1}{2}YX' - u_1\Big(\tbinom{a-1}{3}Y + 4T_3 - E_3^0\Big),\\ B &= \tbinom{a-1}{2}\big(XY' + YX' - 4u_1H_* - 4u_1H_*^+\big) - u_1\Big(\tbinom{a}{3}Y' + \tbinom{a-1}{3}Y + 4T_3 - E_3^0 - \tbinom{a}{3}u_1\Big). \end{aligned}$$

These are forms of the kind treated in step (vi) of the proof of Theorem 5.5: the edge bounds give $A \ge u_1\sum_n e_n^A u_n$ and $B \ge u_1\sum_n e_n^B u_n$. Their coefficients are nonnegative by the same three kinds of certificates: for $4 \le n \le a$ the numerators in (p,k) have nonnegative coefficients and the denominators are strictly positive for $p,k \ge 0$, for $n \le 3$ the polynomials in a have nonnegative coefficients after $a \mapsto a+3$, and for $n \ge a+1$ only nonnegative terms remain (`runs/lean_plan/phi3_generic.py` and `runs/lean_plan/phi3_edge.py` in [7]). Hence $R \ge 0$.

Resummation. Proceed as in the proof of Theorem 4.2, with E in place of W. The inequalities $P^2 \ge 4Q$, $H_+P \ge 10H_-$, $V_+P \ge \frac{9}{2}V_-$, $E_+P \ge 3E_-$, $C_+P \ge 6C_-$ and $A_0P + PA_0 + 2V_1 + E_1 \ge 4H_1$ give $GAG \ge \frac{1}{2}(GA_0 + A_0G) + \frac{1}{18}GV_1G + \frac{1}{12}GE_1G \ge 0$, and (P) follows. ∎

Corollary 5.7 *(Extended cubic ratio).* *Let $0 = d_0 < d_1 < \cdots < d_N = 1$ have Newton coefficients that are log-concave without internal zeros, and let $\rho(x) = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$ with $r > 0$, $t,\tau,\sigma \ge 0$ and $r\sigma \le \frac{7}{4}t\tau$. Put $a_i = 1 - d_i$, $b_i = (1-\rho(i))d_i$ and $c_i = \rho(i)d_i$. If b is nondecreasing, then the system has nonnegative linearization.*

Proof. The proof of Theorem 5.6 uses $r\sigma \le t\tau$ only in (5.5), (5.6) and (5.4), and there only through $\kappa_{t\tau} + \kappa_{r\sigma} \ge 0$; the other coefficients, the key inequality (5.7) and the resummation do not involve the condition. Since

$$t\tau\,\kappa_{t\tau} + r\sigma\,\kappa_{r\sigma} \;=\; \big(t\tau - \tfrac{4}{7}r\sigma\big)\kappa_{t\tau} + \tfrac{4}{7}r\sigma\big(\kappa_{t\tau} + \tfrac{7}{4}\kappa_{r\sigma}\big),$$

under $r\sigma \le \frac{7}{4}t\tau$ it suffices that $\kappa_{t\tau} \ge 0$ and $\kappa_{t\tau} + \frac{7}{4}\kappa_{r\sigma} \ge 0$ as forms in u. For (5.4), for (5.5) with $n \ge 6$ and for (5.6), where $u_0 = 0$, the tail sums of the modified coefficient $\kappa_{t\tau} + \frac{7}{4}\kappa_{r\sigma}$ stay nonnegative. For (5.5) with $n \le 5$ the transfers alone do not suffice. There the modified coefficient is certified after the substitution $u_0 = zu_1$, $0 \le z \le \min(\ell, 1/q)$ with $q = u_2/u_1$, in the regimes $z = 0$, $z = \ell$ and $z = 1/q$ (`runs/cubic/bdetour_u.py` in [7]; `runs/cubic/edetour_u.py` certifies (5.4) independently in the same way). The relevant expressions are affine in z, so these boundary regimes cover the admissible interval for $q > 0$. When $q = 0$, all u_k with $k \ge 2$ vanish, and the inequality follows directly. ∎

The Lean proof, `Szkraw.cubic_ratio74_matrix`, proves Corollary 5.7 directly and derives Theorem 5.6 from it. For (5.5) with $n \le 5$ it does not substitute $u_0 = zu_1$: it writes the modified coefficient, times a positive polynomial in ℓ, as a combination of transfers, products $u_a u_b$ and the terms $u_j(\ell u_1 - u_0) \ge 0$, with multipliers that are polynomials in ℓ with nonnegative coefficients after a shift of ℓ, plus exact rational certificates at the few smaller values of ℓ (`runs/lean_plan/cert74.py` and `runs/lean_plan/gen_cubicB.py` in [7]; see the supplement [9]). Exact diagnostics for both routes are listed in the documentation [8].

Remark 5.8. The factor $\frac{7}{4}$ is a limit of the certificates, not of property (P). On geometric profiles at $(\ell, n) = (0, 3)$ one has $\kappa_{r\sigma} = -20q(4q^3 + 11q^2 + 9q + 1)$ and $\kappa_{t\tau} = 20(q+1)(8q^3 + 33q^2 + 30q + 6)$ in (5.5), so $\kappa_{t\tau}/|\kappa_{r\sigma}| \to 2$ as $q \to \infty$. This family therefore obstructs the coefficientwise test for $r\sigma \le u\,t\tau$ with $u > 2$; it is not a counterexample to property (P). With $u = \frac{9}{5}$ the certificate of `runs/cubic/bdetour_u.py` fails. Whether $r\sigma \le u\,t\tau$ suffices for some $u > \frac{7}{4}$ is open (Section 6).

6. Open problems

The preceding criteria cover several increasing ratios, but simple shape restrictions alone do not imply property (P). Exact counterexamples show that log-concavity of the Newton coefficients of both d and c with nondecreasing b, a merely increasing ratio with nondecreasing b, and nonnegative Newton coefficients of a cubic ratio can each fail. The parameters and negative product coefficients are collected in Appendix A, separately from sampled tests, which establish no threshold.

Find direct inequalities for other varying ratios, especially those outside Theorems 3.1, 4.1 and 5.5. Determine whether the cubic bound of Corollary 5.7 can be enlarged, and whether the log-concavity assumption on ratios of degree greater than three can be weakened to conditions like the one in Theorem 5.6. The coefficientwise detour test fails on geometric profiles beyond the factor 2 (Remark 5.8), which is a limit of that test rather than a counterexample to property (P). Find a proof of the key inequality (5.2)

that does not use computer certificates. All these questions include the terminal row and products whose degree sum exceeds N.

A. Exact counterexamples and sampled tests

All examples in this appendix are reproduced in the companion code [7]. The counterexamples of the first paragraph are exact and machine-verified. The tests of the second paragraph are sampled numerical evidence; they are not formalized.

Exact counterexamples. Each array below satisfies (2.1) and has a negative product coefficient.

- *Log-concave Newton coefficients of d and c with nondecreasing b.* The arrays $d = (0, 25, 55, 91)/91$ and $c = (0, 24, 54, 90)/91$ give $g(3, 3; 3) = -3/2002$.
- *An increasing concave ratio with nondecreasing b.* The linear profile $d_i = i/3$ with $\rho = (1/2, 3/4, 5/6)$, for which $b_i = 1/6$ for $i \geq 1$, gives $g(3, 3; 3) = -1/24$.
- *A cubic ratio with nonnegative Newton coefficients.* For $N = 4$, $d_i = i/4$ and $\rho(x) = \frac{19}{20} + \binom{x}{3}/320$, whose Newton coefficients $(\frac{19}{20}, 0, 0, \frac{1}{320})$ are nonnegative but have internal zeros, $b = (0, \frac{1}{80}, \frac{1}{40}, \frac{9}{256}, \frac{3}{80})$ is nondecreasing and $g(4, 4; 4) = -14011/30720000$. So Theorem 4.2 does not extend to cubic ratios without further conditions.
- *Cubic ratios with $r \leq \frac{9}{10}$.* On linear profiles with $\kappa = 1/80$, which keeps b nondecreasing, $\rho_i = \frac{9}{10} + \kappa\binom{i}{3}/\binom{N}{3}$ gives $g(48, 48; 48) < 0$ for $N = 48$, and $\rho_i = \frac{9}{10} + \kappa(i/N)^3$ gives $g(64, 64; 63) < 0$ for $N = 64$.
- *Steep ratios.* On linear profiles, with nondecreasing b and nonnegative Newton coefficients of ρ, the ratio $\rho_i = \frac{1}{2} + 10^{i-N}/100$ gives $g(40, 40; 40) < 0$ for $N = 40$. Raising only the last ratio, $\rho_i = r$ for $i < N$ and $\rho_N = r + (1 - r)/N$, gives $g(40, 40; 38) < 0$ for $r = \frac{1}{2}$, $N = 40$, and $g(120, 120; 119) < 0$ for $r = \frac{2}{5}$, $N = 120$; these two ratios are convex, with a single kink.

Write a cubic ratio as $\rho(x) = r + tx + \tau\binom{x}{2} + \sigma\binom{x}{3}$. The cubic examples violate $r\sigma \leq t\tau$, a consequence of log-concavity without internal zeros of (r, t, τ, σ): in the first two $t = \tau = 0$, and for $\kappa(i/N)^3$ one has $t\tau = 6\kappa^2/N^6 < r\sigma = 6r\kappa/N^3$. One further exact computation concerns the method rather than property (P): for $t = \tau = 0$ the operator M of the proof of Theorem 4.1 can have negative entries, for instance for $N = 6$, $d_i = i/6$, $r = \frac{1}{5}$ and $\sigma = \frac{1}{200}$. This obstructs extending that resummation to such data, and does not contradict Theorem 4.1, which does not cover them.

Sampled tests without failures. These are numerical evidence over finitely many arrays, not certified thresholds. In sampled arrays with $t, \tau \geq 0$, $r\sigma = u\, t\tau$ and nondecreasing b on log-concave Newton profiles, no negative coefficients were found for $u \in \{\frac{1}{4}, 1, 4, 16, 64\}$, $r \in \{\frac{1}{10}, \frac{1}{3}, \frac{1}{2}, \frac{4}{5}, \frac{19}{20}, \frac{99}{100}\}$ and $4 \leq N \leq 18$ (full tables), or in the coefficients $g(m, N; k)$ for

$u \in \{1, 4, 16\}$ and $N = 24, 32, 48, 64, 96$. In these samples the operator M of the proof of Theorem 4.1 was nonnegative for $u \le 4$ and $N \le 13$. The samples include values $u > \frac{7}{4}$, so they bear on the question of Section 6 whether the bound of Corollary 5.7 can be enlarged, but they do not decide it.

B. Proof of Proposition 5.3

We recall the statement: if $(\rho_i)_{i\ge 0}$ and $(\nu_j)_{j\ge 0}$ are nonnegative and log-concave without internal zeros, $f(x) = \sum_i \rho_i \binom{x}{i}$ and $g(x) = \sum_j \nu_j \binom{x}{j}$, then the Newton coefficients $\kappa_k = \Delta^k(fg)(0)$ of fg are nonnegative and log-concave without internal zeros.

Proof. The product rule $\binom{x}{i}\binom{x}{j} = \sum_k \binom{k}{i}\binom{i}{k-j}\binom{x}{k}$ gives $\kappa_k = \sum_p H_k(p)\rho_{p_1}\nu_{p_2}$, where $H_k(i,j) = \binom{k}{i}\binom{i}{k-j} = [y^i w^j](y + w + yw)^k$ vanishes unless $i, j \le k \le i + j$. If either sequence vanishes identically, so does κ. Otherwise, if ρ and ν are supported on $[p_0, p_1]$ and $[q_0, q_1]$, with $p_1, q_1 \le \infty$, then $\kappa_k > 0$ exactly for $\max(p_0, q_0) \le k \le p_1 + q_1$, so κ has no internal zeros. For log-concavity fix $k \ge 1$; then

$$\kappa_k^2 - \kappa_{k-1}\kappa_{k+1} = \sum_{c\in\mathbb{Z}^2} \sum_{p\in\mathbb{Z}^2} T_c(p)\, u_c(p_1)\, w_c(p_2),$$
$$T_c(p) = H_k(p)H_k(c-p) - H_{k-1}(p)H_{k+1}(c-p),$$

with $u_c(i) = \rho_i\rho_{c_1-i}$ and $w_c(j) = \nu_j\nu_{c_2-j}$. By log-concavity without internal zeros, u_c is symmetric about $c_1/2$ and nondecreasing on $i \le c_1/2$, so $u_c = \sum_{0\le a\le c_1/2}(u_c(a) - u_c(a-1))\mathbf{1}_{[a,c_1-a]}$ with nonnegative coefficients, and likewise w_c. It therefore suffices to show $D_c(B) = \sum_{p\in B} T_c(p) \ge 0$ for every box $B = [a, c_1 - a] \times [b, c_2 - b]$ with $a \le c_1/2$, $b \le c_2/2$.

Let $E = \{(1,0), (0,1), (1,1)\}$, so that $H_k = \sum_{e\in E} H_{k-1}(\cdot - e)$. Expanding $H_k(p)$, $H_k(c-p)$ and $H_{k+1}(c-p)$ in this way gives

$$D_c(B) = \sum_{e,e'\in E} \big(V_{e+e'}(B-e) - V_{e+e'}(B)\big), \qquad V_s(p) = H_{k-1}(p)H_{k-1}(c-s-p),$$

where $V_s(X) = \sum_{p\in X} V_s(p)$. Fix $s = e + e'$, write $V = V_s$, $m = c - s$, $t = k - 1$ and $\mu = m/2$. With $n_1 = 2t - m_2$, $n_2 = 2t - m_1$, $n_3 = m_1 + m_2 - 2t$ one has $V(x,y) = \frac{t!^2}{n_1!n_2!n_3!}\binom{n_1}{t-y}\binom{n_2}{t-x}\binom{n_3}{x+y-t}$ (and $V = 0$ if some $n_i < 0$). Three properties follow.

(T1) For every interval W, the marginal $G_W(x) = \sum_{y\in W} V(x,y)$ is a constant times $\binom{n_2}{t-x}\sum_{z\in t-W}\binom{n_1}{z}\binom{n_3}{x-z}$, a product of a binomial row and a convolution of two log-concave sequences without internal zeros; so G_W is log-concave without internal zeros. The same holds with the roles of the coordinates exchanged.

(T2) $V(c-s-p) = V(p)$, and V is reverse regular of order two: $V(x,y)V(x',y') \le V(x,y')V(x',y)$ for $x \le x'$, $y \le y'$, by log-concavity of $\binom{n_3}{\cdot}$. Taking $x' = 2\mu_1 - x$ and y, y' symmetric about μ_2, central symmetry turns this into $V(x,y)^2 \le V(x,y')^2$. Hence $V(x,y') \ge V(x,y)$ whenever $x \le \mu_1$, $y \le y'$ and $y + y' = 2\mu_2$; and symmetrically in the coordinates.

(T3) If G is log-concave without internal zeros and $G(x) \le G(2\mu_1 - x)$ for all $x \le \mu_1$, then G is nondecreasing on $x \le \mu_1$: otherwise $G(x-1) > G(x)$ for some $x \le \mu_1$, and log-concavity makes G nonincreasing from $x-1$ on, so $G(2\mu_1 - x + 1) \le G(x) < G(x-1)$.

Write $I = [a, c_1 - a]$, $J = [b, c_2 - b]$, $X(I', J') = V(I' \times J')$, and telescope along rows or columns. By the symmetry $H_k(i,j) = H_k(j,i)$ it suffices to treat five ordered pairs.

$e = e' = (1,0)$: $\mu = c/2 - (1,0)$ and $V(B-e) - V(B) = G_J(a-1) - G_J(a-2)$ by central symmetry. Since J is symmetric about μ_2, G_J is symmetric about μ_1, hence nondecreasing on $x \le \mu_1$ by (T1) and (T3); and $a - 1 \le \mu_1$.

$e = (1,0)$, $e' = (0,1)$: $\mu = (c - (1,1))/2$ and $V(B-e) - V(B) = V(a-1, c_2 - b) - V(a-1, b-1) \ge 0$ by (T2), since $a - 1 < \mu_1$ and $(c_2 - b) + (b-1) = 2\mu_2$.

$e = (1,0)$, $e' = (1,1)$: $\mu = c/2 - (1, \frac{1}{2})$ and

$$V(B-e) - V(B) = \big[V(a-1, c_2 - b) - V(a-1, b-1)\big] + \big[G_{J-1}(a-1) - G_{J-1}(a-2)\big].$$

The first bracket is nonnegative by (T2), as $a - 1 \le \mu_1$. For the second, $G_{J-1}(2\mu_1 - x) - G_{J-1}(x) = V(x, c_2 - b) - V(x, b-1) \ge 0$ for $x \le \mu_1$ by (T2), so (T1) and (T3) apply.

$e = (1,1)$, $e' = (1,0)$: the same V, and $V(B-e) - V(B) = \big[X(I-1, J-1) - X(I-1, J)\big] + \big[V(B - (1,0)) - V(B)\big]$. The first bracket vanishes by central symmetry, since $I - 1$ is symmetric about μ_1; the second is the previous case.

$e = e' = (1,1)$: $\mu = c/2 - (1,1)$ and

$$\begin{aligned} V(B-e) - V(B) = {} & \big[K(b-1) - K(b-2)\big] + \big[X(a-1, J) - X(a-1, J-2)\big] \\ & + \big[G_{J-2}(a-1) - G_{J-2}(a-2)\big] \end{aligned}$$

with $K(y) = \sum_{x \in I-1} V(x,y)$. Since $I - 1$ is symmetric about μ_1, K is symmetric about μ_2 and nondecreasing on $y \le \mu_2$ by (T1) and (T3); and $b - 1 \le \mu_2$. The second bracket is $[V(a-1, c_2 - b) - V(a-1, b-2)] + [V(a-1, c_2 - b - 1) - V(a-1, b-1)]$, two differences of the form in (T2), since both pairs of second coordinates sum to $2\mu_2 = c_2 - 2$. For the third, $G_{J-2}(2\mu_1 - x) - G_{J-2}(x) = X(x, J) - X(x, J-2) \ge 0$ for $x \le \mu_1$ by (T2), so (T1) and (T3) give $G_{J-2}(a-1) \ge G_{J-2}(a-2)$.

Each of the nine ordered-step contributions $V_{e+e'}(B-e) - V_{e+e'}(B)$ is nonnegative, so $D_c(B) \ge 0$, which proves the proposition. (The individual $T_c(p)$ need not be nonnegative.) ■

Diagnostics in the companion code [7] check in exact arithmetic the identity for $D_c(B)$, the sign of each of its nine terms for $k \le 9$, and every bracket of the case analysis, (T1) and (T2) for $k \le 6$.

AI-use disclosure

Claude Opus 5.5 and, for parts of the Lean formalization of Theorem 5.5, Claude Haiku (Anthropic; the Haiku version was not recorded), both run in the Claude Code agent; GPT-6 Astra or GPT-6.1 Sol (OpenAI; which of the two was not recorded), run in the Codex

agent; and GPT-6 Pro (OpenAI), used in ChatGPT, were used to develop the proofs, the exact checks and certificates, the Lean formalization and the text. The resummation and the key inequalities of Theorems 4.1 and 4.2 arose in consultations with GPT-6 Pro, which reviewed every result; Theorems 5.1, 5.5 and 5.6 were found with Claude Opus 5.5. This AI review is not human peer review, and no independent human verification is claimed. The author set the research objective and is responsible for the mathematical claims. The full disclosure is in the supplement [9].

Code and formalization

All theorems, corollaries and propositions, and Lemma 5.4, are formalized in Lean 4 with Mathlib in the companion code [7], whose axiom audit admits only `propext`, `Classical.choice` and `Quot.sound`; "machine-verified" refers only to these Lean theorems. The certificate scripts were also run unmodified from a fresh clone at the cited revision [8]. The supplement [9] lists the Lean declaration for each result and remark, gives the inventory of computer-assisted steps with their ranges, scripts and Lean declarations, and lists what is not formalized, chiefly the sampled tests of Appendix A and the comparisons with the literature.

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